

# From Trees to Networks



ISMIB 2026 Tutorial IP3

Daniel H. Huson, Anupam Gautam and Banu Cetinkaya

# Your presenters today



Daniel Huson

- Professor of Algorithms in Bioinformatics
- University of Tübingen
- Phylogenetic networks, SplitsTree, PhyloSketch, PhyloCompare



Anupam Gautam

- Post-Doc
- University of Tübingen
- Max-Planck Institute for Biology Tübingen
- PhyloGuide, AI-assisted phylogenetic analysis



Banu Cetinkaya

- PhD student
- University of Tübingen
- Phylogenetic networks, co-developer of PhyloCompare

# What we'll do today (09:00 to 13:00)

## 09:00 - 10:50 Trees

- Per-locus inference with IQ-TREE and BEAST X
- Coalescent species tree with ASTRAL

➤ “We have many gene trees and they disagree”

## 10:50 - 11:10 / Break ☕

## 11:10 - 13:00 Networks

- SplitsTree & Neighbor-Net on alignments
- PhyloSketch for interactive network construction
- PhyloCompare for networks from gene tree sets



# Agenda

1. Tutorial overview and dataset
2. Setup verification
3. **Multi-locus phylogenomics lecture**
4. IQ-TREE 3 hands-on
5. BEAST X + SplitsTree posterior visualization
6. ASTRAL hands-on 
7. **Networks lecture**
8. SplitsTree Neighbor-Net hands-on
9. PhyloSketch hands-on
10. PhyloCompare hands-on
11. Biological interpretation + wrap-up





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# Dataset: the *Anopheles gambiae* complex

## RESEARCH ARTICLE

### MOSQUITO GENOMICS

## Extensive introgression in a malaria vector species complex revealed by phylogenomics

Michael C. Fontaine,<sup>1,2\*†</sup> James B. Pease,<sup>3\*</sup> Aaron Steele,<sup>4</sup> Robert M. Waterhouse,<sup>5,6,7,8</sup> Daniel E. Neafsey,<sup>6</sup> Igor V. Sharakhov,<sup>9,10</sup> Xiaofang Jiang,<sup>10</sup> Andrew B. Hall,<sup>10</sup> Flaminia Catteruccia,<sup>11,12</sup> Evdoxia Kakani,<sup>11,12</sup> Sara N. Mitchell,<sup>11</sup> Yi-Chieh Wu,<sup>5</sup> Hilary A. Smith,<sup>1,2</sup> R. Rebecca Love,<sup>1,2</sup> Mara K. Lawnczak,<sup>13†</sup> Michel A. Slotman,<sup>14</sup> Scott J. Emrich,<sup>2,4</sup> Matthew W. Hahn,<sup>3,15§</sup> Nora J. Besansky<sup>1,2§</sup>

*Science* **347** (6217), 1258524. DOI: 10.1126/science.1258524

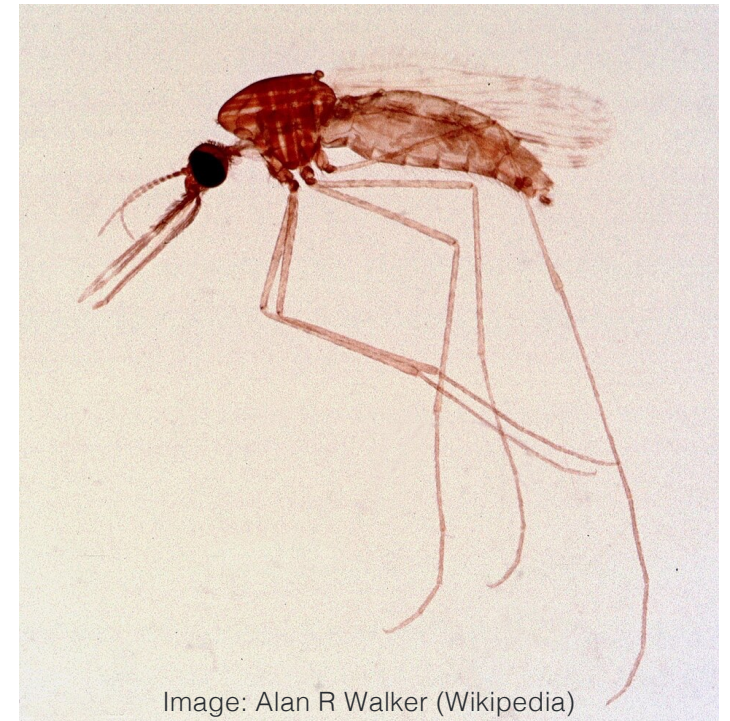
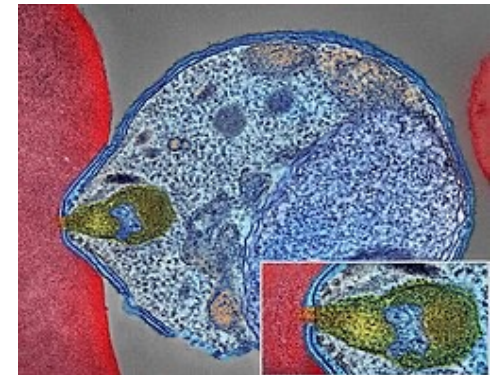


Image: Alan R Walker (Wikipedia)

# Dataset: the *Anopheles gambiae* complex

Why public health cares:

- Three of six species are the principal vectors of *P. falciparum* malaria
- How did they acquire vectorial capacity?
- Understanding the phylogeny matters for understanding the disease

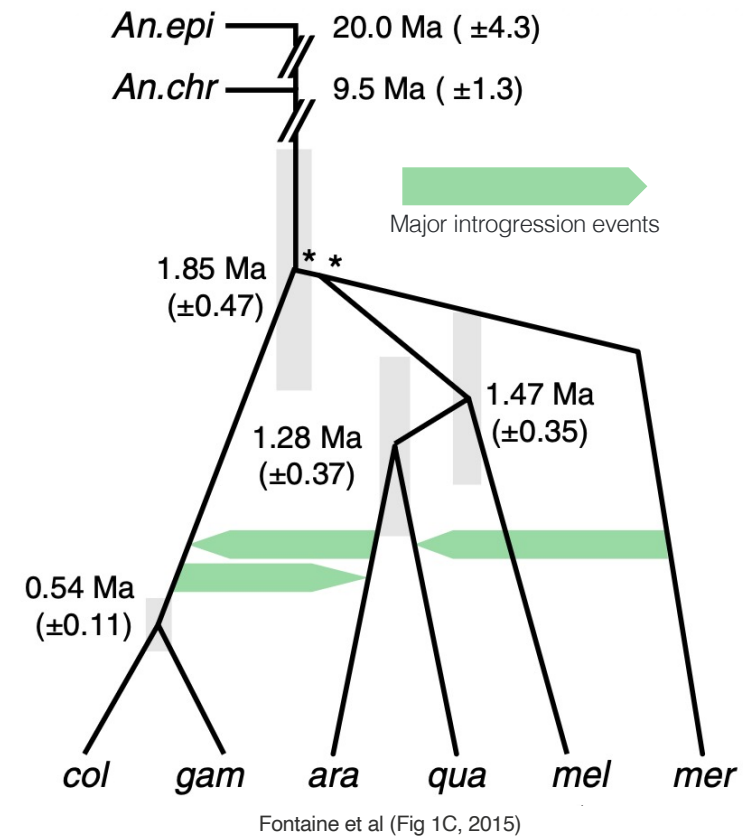


Malaria parasite connecting to a red blood cell  
NIAID, Wikipedia

# Dataset: the *Anopheles gambiae* complex

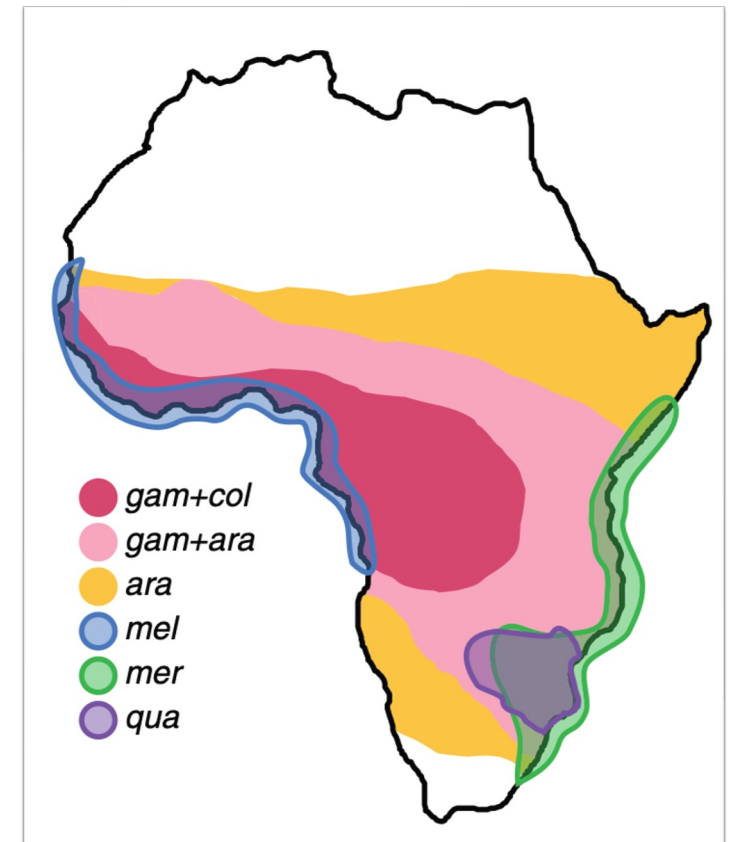
## Why phylogenetics cares:

- Six closely related species, recently diverged (about 2 Mya)
- Pervasive autosomal introgression
- X chromosome tells a different story than autosomes
- Several different discordance signals



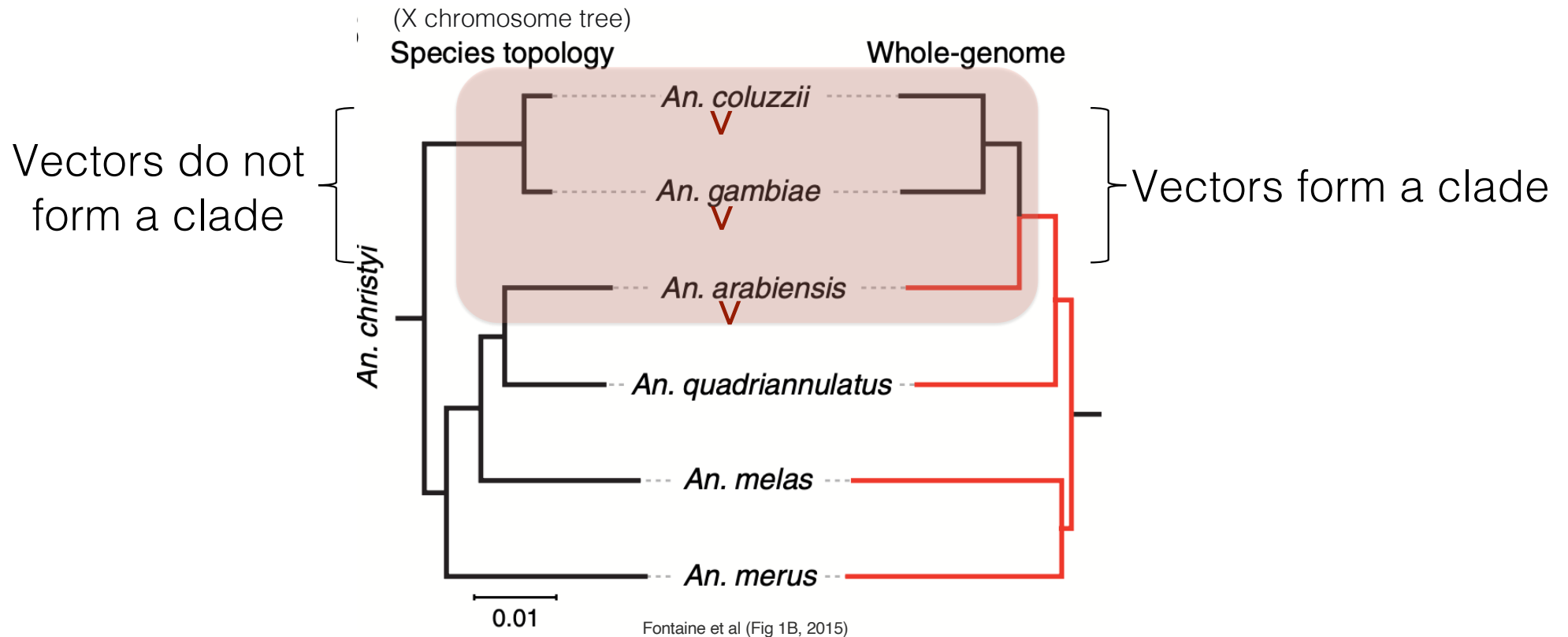
# Six species, one complex

- *Anopheles gambiae* (vector)
- *An. coluzzii* (vector)
- *An. arabiensis* (vector)
- *An. quadriannulatus* (non-vector)
- *An. melas* (minor vector, brackish water)
- *An. merus* (minor vector, brackish water)
- *An. christyi* (outgroup)



Fontaine et al (Fig 1A, 2015)

# The same dataset, two different trees



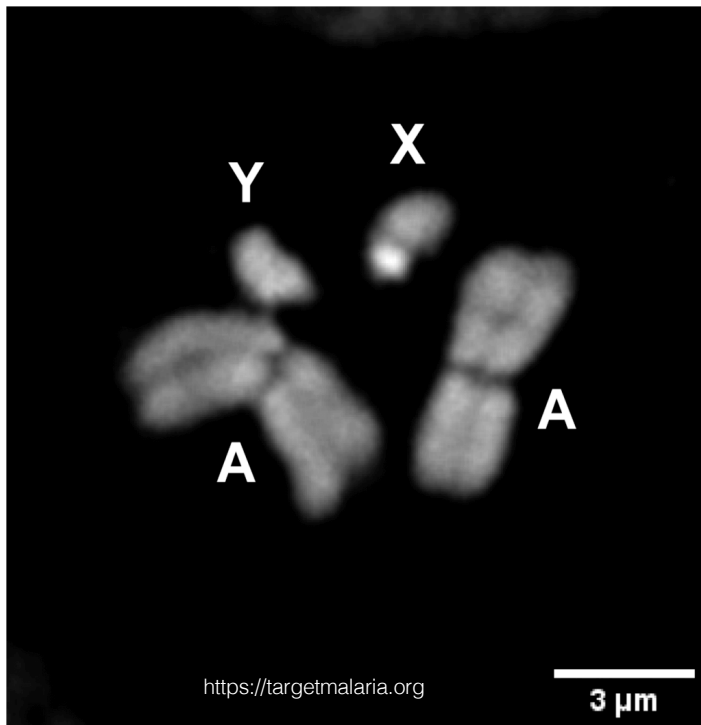
- Both trees 100% bootstrap support, which is “correct”?

# The X chromosome tells the truth

Key findings (Fontaine et al, 2015):

- 1. The X chromosome tree is the species tree.**  
Higher sequence divergence between sister pairs confirms this is the historical branching order.
- 2. The autosomes have been pervasively introgressed**  
between *An. arabiensis* and (*An. gambiae* + *An. coluzzii*).
- 3. Vectorial capacity may have spread through**  
introgression, not only through de novo mutation.

# Our data: 15 loci sampled to show the story



| Region                      | Loci | Expected topology                                      |
|-----------------------------|------|--------------------------------------------------------|
| X distal<br>(Xag inversion) | 5    | (arabensis, quadriannulatus) -<br>species tree         |
| X pericentromeric           | 1    | (arabensis, (gambiae, coluzzii)) -<br>introgressed     |
| Autosomal (2R, 3R)          | 5    | (arabensis, (gambiae, coluzzii)) -<br>introgressed     |
| 2La inversion (2L)          | 2    | 2La-specific topologies                                |
| 3La inversion (3L)          | 2    | (merus, quadriannulatus) -<br>unexpected introgression |

Names such as 2La or 3La refer to specific chromosomal inversions



# Software for trees and networks

## Part I (trees)

IQ-TREE 3

per-locus ML

BEAST X

Bayesian inference

- BEAUti
- Tracer,
- TreeAnnotator

ASTRAL

MSC species tree

## Part II (networks)

SplitsTree

Consensus networks,  
Neighbor-Net

PhyloSketch

interactive sketching

PhyloCompare

gene-trees-to-network

## Throughout

- PhyloGuide

AI assistant



# What we will do today

When the genome has been pervasively introgressed, tree methods can return confident but wrong answers. Networks can make the conflict visible.

- In Part I, we will infer gene trees and a species tree from a multi-locus alignment.
  - We will see that the trees disagree systematically across the genome.
- In Part II, we will use phylogenetic networks to visualize and interpret that disagreement.



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# Quick setup verification

☐ Repository cloned:

```
git clone https://github.com/husonlab/trees-to-networks-tutorial.git
```

☐ `iqtree3 --version` works

☐ `java -jar /path/to/astral.5.7.8.jar -h` works

☐ `beauti` launches

☐ `SplitsTree` launches

☐ `tracer` launches

☐ `PhyloSketch` launches

☐ `PhyloCompare` launches

☐ `PhyloGuide` URL bookmarked

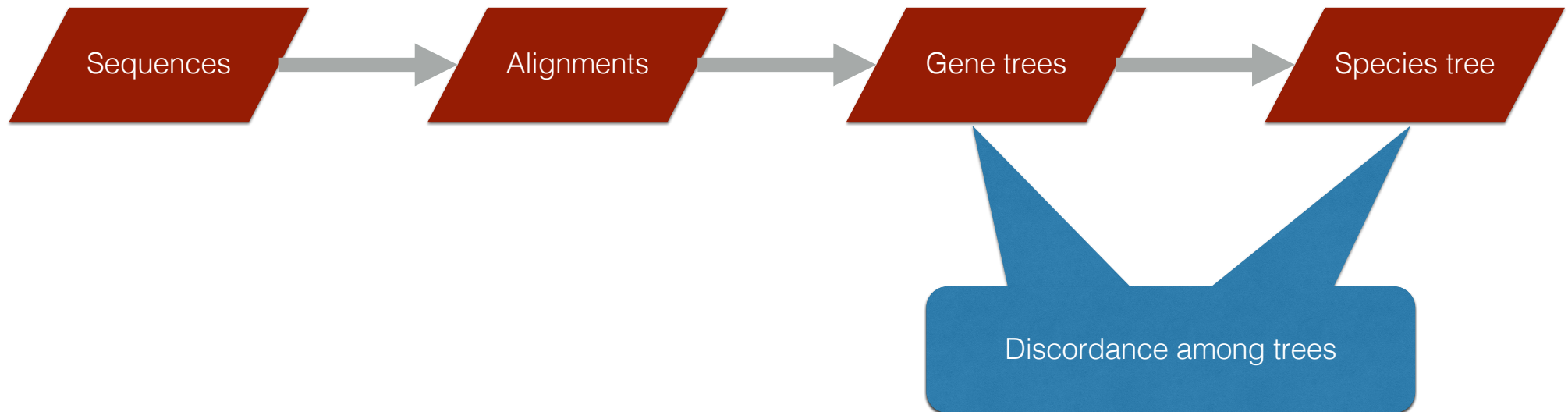


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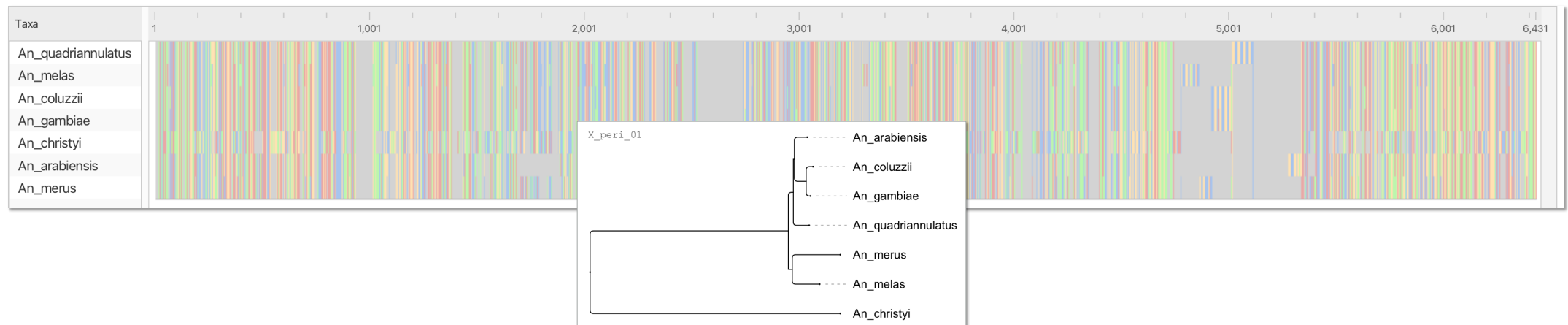


# From sequences to gene trees to species trees



# At a single locus

- A multiple sequence alignment (MSA) is our raw input
- 7 species, ~5000 nucleotide positions, every column is informative about how the species relate
- From the MSA we infer a **gene tree**: the evolutionary history of this one stretch of DNA



# Inferring a tree by maximum likelihood

- **Likelihood:** probability of observing the alignment  $A$  given tree  $T$  and substitution model  $M$ :

$$P(A \mid T, M)$$

- Find the tree and model that maximize this probability
- **Tool: IQ-TREE 3**
- Substitution models (JC, K2P, HKY, GTR, +G, +I)  
how nucleotides change over time
- Bootstrap: assess confidence in the tree (we use UFBoot)



# Inferring a *distribution* of trees by Bayesian inference

- Bayesian inference returns the full posterior distribution
- Not just one best tree but  $P(T, M | A)$  for "every" possible tree
- Tool: **BEAST X** (uses MCMC sampling)
- Thousands of trees, each weighted by posterior probability
- Captures uncertainty natively

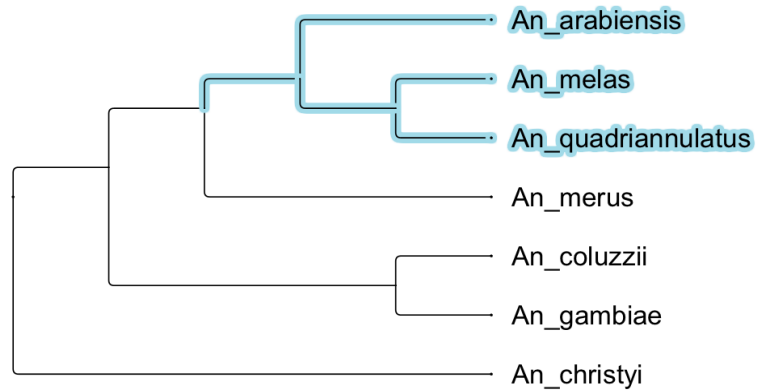
# Genome-scale data: many loci, many gene trees

## Phylogenomics:

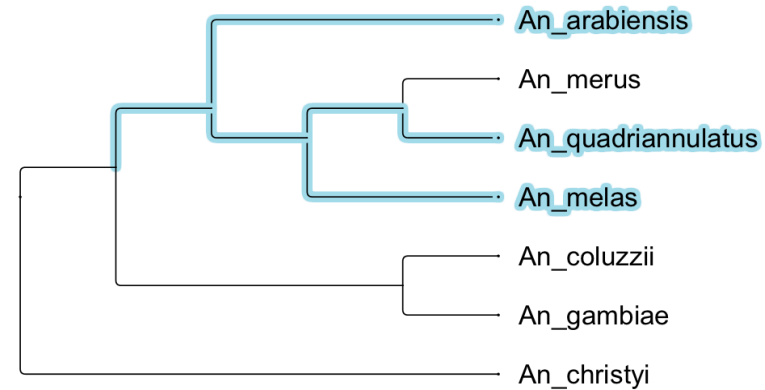
- hundreds to thousands of loci per analysis
- Each locus has its own evolutionary history
- Run ML or Bayesian inference per locus -> a *set* of gene trees
- For our tutorial: **15 loci, 15 gene trees**

# Gene trees disagree

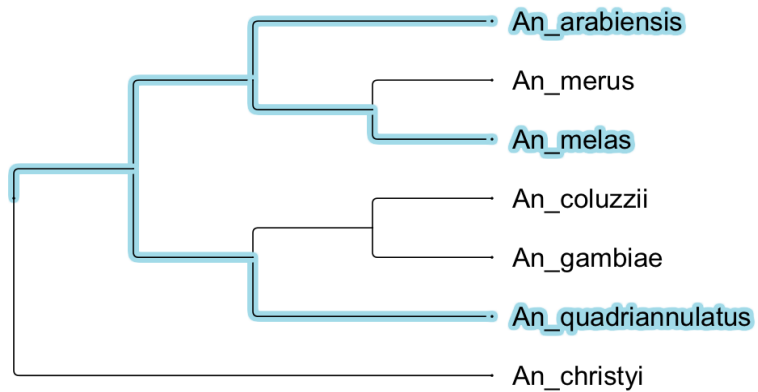
X\_dist\_01



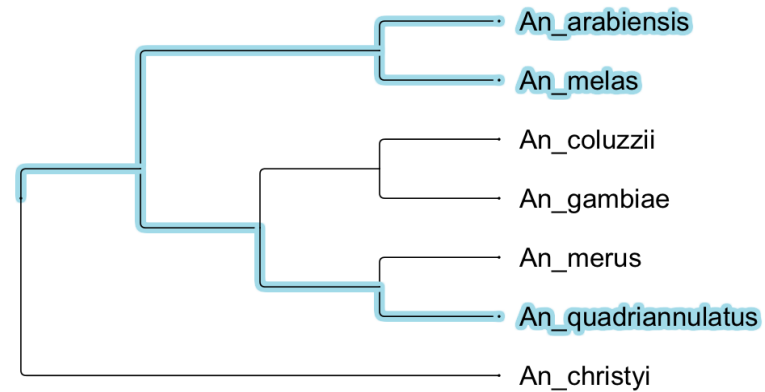
auto\_2R\_01



inv\_2La\_02

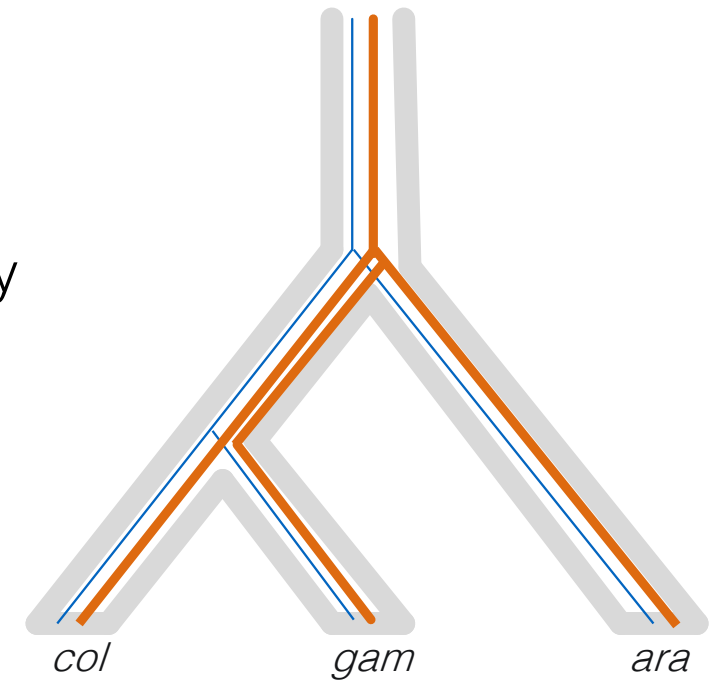


inv\_3La\_01



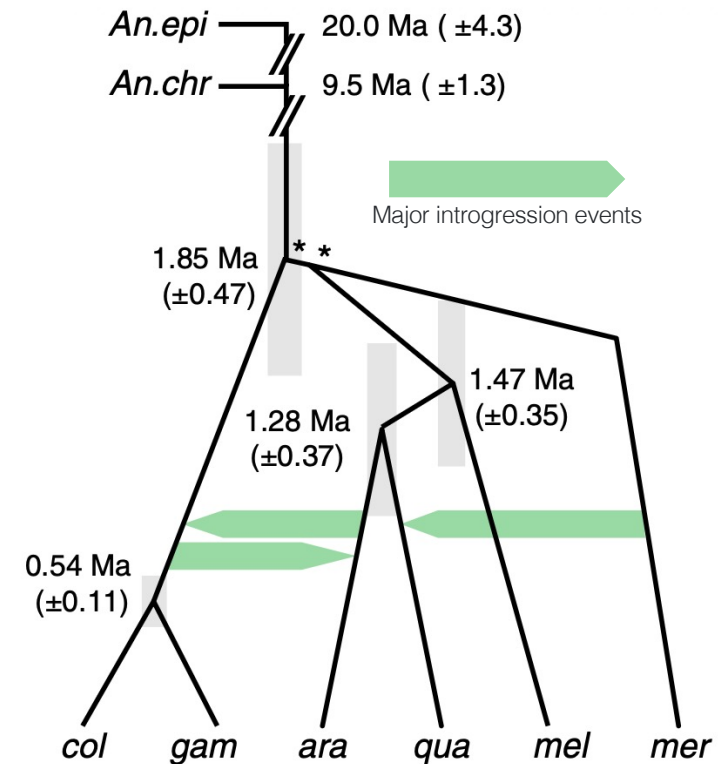
## Cause #1 of discordance: incomplete lineage sorting (ILS)

- Ancestral polymorphism can persist across multiple species
- Different alleles get inherited by different descendant lineages
- The gene tree can have a different topology than the species tree
- Purely stochastic; no gene flow required



## Cause #2 of discordance: introgression

- After speciation, gene flow between species can re-introduce alleles
- Hybridization, horizontal transfer
- Introgression is a **biological** process  
not just stochastic
- Introgressed alleles look like the donor species, not like closest relatives
- Often non-uniform across the genome  
(only some regions introgressed)



Fontaine et al (Fig 1C, 2015)

# Telling ILS and introgression apart

- Both ILS and introgression produce gene trees that disagree with the species tree
- ILS: the alleles that don't sort are **ancestral**, so they coalesce **deep** in the species tree
- Introgression: the alleles came from a sister species **recently**, so they coalesce **shallow**
- This is how Fontaine et al. (2015) showed *Anopheles* autosomes are introgressed, not just ILS

# Now: let's try to see this in our data

- 15 loci spanning autosomes, X chromosome, and inversion regions
- Run IQ-TREE on each -> 15 gene trees
- We expect to see discordance
- Question: where does the discordance occur?

| Region                      | Loci | Expected topology                                      |
|-----------------------------|------|--------------------------------------------------------|
| X distal<br>(Xag inversion) | 5    | (arabiensis, quadriannulatus) -<br>species tree        |
| X pericentromeric           | 1    | (arabiensis, (gambiae, coluzzii)) -<br>introgressed    |
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# PhyloGuide Q1

**Q:** We have multi-locus alignments from a recently diverged species complex. How should we infer a species phylogeny?

**A:** Infer gene trees for each locus, then estimate a species tree with ASTRAL instead of relying only on concatenation. This accounts for gene-tree discordance caused by ILS.







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# Hands-on: per-locus ML inference with IQ-TREE 3

- Time budget: ~20 minutes
- Goal: produce 15 maximum-likelihood gene trees, one per locus, and see how they vary across the genome
- Where we are: `cd trees-to-networks-tutorial`
- What we'll run: `iqtree3 -S alignments/ ...`  
(full command on next slide)



# The IQ-TREE 3 command

```
iqtree3 -S data/alignments -B 1000 --prefix tutorial_loci -T AUTO -st DNA
```

| Flag                   | What it does                                         |
|------------------------|------------------------------------------------------|
| -S alignments          | Per-locus mode: one tree per FASTA in this directory |
| -B 1000                | Ultrafast bootstrap with 1000 replicates             |
| --prefix tutorial_loci | Output filename prefix                               |
| -T AUTO                | Auto-pick number of threads                          |
| -st DNA                | Force DNA mode (auto-detection fails on some loci)   |

# Run the IQ-TREE 3 command

Run the command.

- What runs internally:
  - ModelFinder picks the best substitution model per locus
  - then maximum-likelihood tree search
  - then UFBoot for branch support
- Runtime: ~ 5 seconds
- Output file: `tutorial_loci.treefile`

# Run the IQ-TREE 3 command

- You'll see one block of output per locus, ending like this:

```
Subset  Type   Seqs   Sites  Infor  Invar  Model  Name
1      DNA     7   6208   144    5068      X_dist_01.fasta
2      DNA     7   7530   234    5899      X_dist_02.fasta
...

15     DNA     7   5454 105  4455      inv_3La_02.fasta
```

- What's normal:
  - All 15 loci have 7 sequences
  - Most loci pick HKY+F+G4 as best-fit model
  - Christyi (the outgroup) failing composition chi2 tests in several loci is not a problem

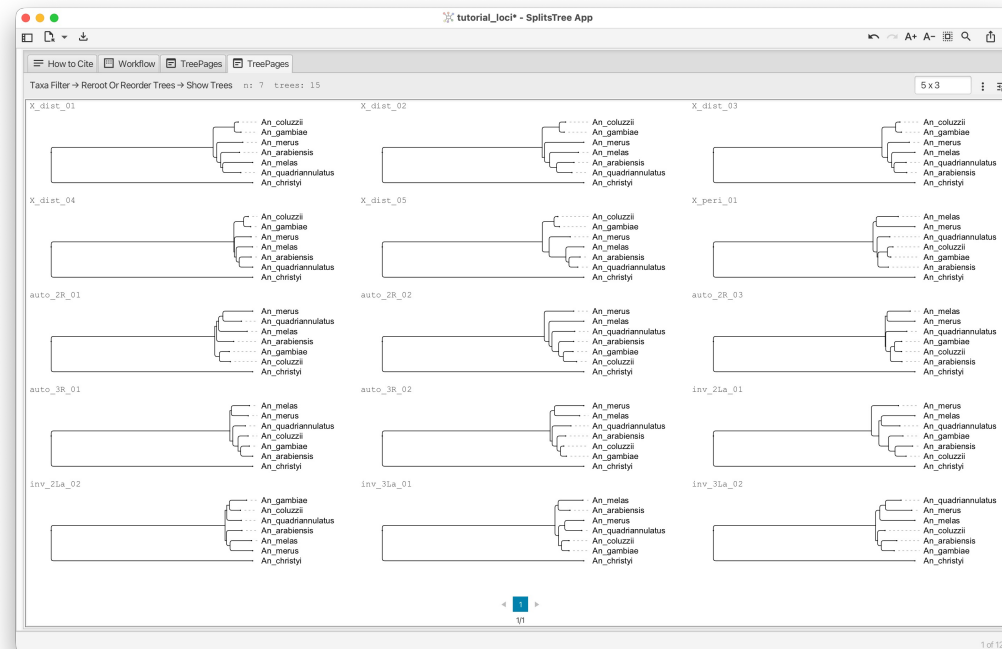
## Nexus file for SplitsTree and PhyloCompare

- IQ-TREE wrote 15 Newick trees to `tutorial_loci.treefile`, one per line in alphabetical order of FASTA filenames
- For SplitsTree and PhyloCompare, we want a NEXUS file with each tree labeled by its locus ID and correctly rooted by outgroup `An_christyi`
- Run our helper:

```
python data-prep/make_nexus_treeset.py \  
    --alignments-dir alignments/ \  
    --treefile tutorial_loci.treefile \  
    --out tutorial_loci.nex\  
    --outgroup An_christyi
```

# Open the gene trees in SplitsTree and look around

- Open `tutorial_loci.nex` in SplitsTree
  - File -> Open -> select the .nex file
- What you should see: 15 named trees in the tree set



# Discussion items

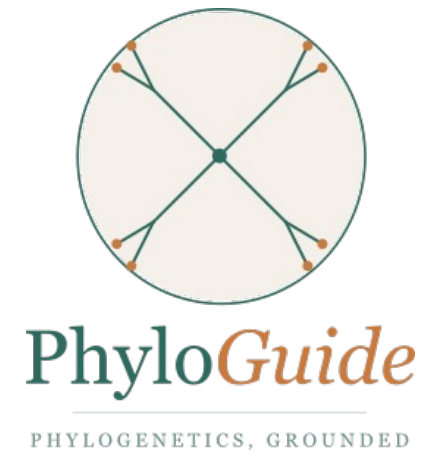
- Do all loci show the same topology? Why not?
- Can you tell X-distal loci apart from autosomal ones just by topology?
- Where in the genome would you expect to see "species-tree" signal vs "introgressed" signal, based on what we know about this dataset?



# PhyloGuide Q2

**Q:** We ran IQ-TREE on 15 alignments and the gene trees disagree across chromosomes - X chromosome shows one topology, autosomes another. What does this mean?

**A:** Different genomic regions can have different evolutionary histories due to ILS, introgression, selection, or gene-tree error. It does not mean one topology is necessarily correct.





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# Hands-on: Bayesian inference + visualizing the posterior

- **Time budget:** 30 minutes
- **Goal:** produce a posterior tree distribution for one locus, then visualize the within-locus uncertainty as a network
- **Tools:** BEAST X, BEAUti, Tracer, TreeAnnotator, SplitsTree
- **Locus:** X\_dist\_04 (one of our cleanest X-distal alignments)



# From “best tree” to “distribution of trees”

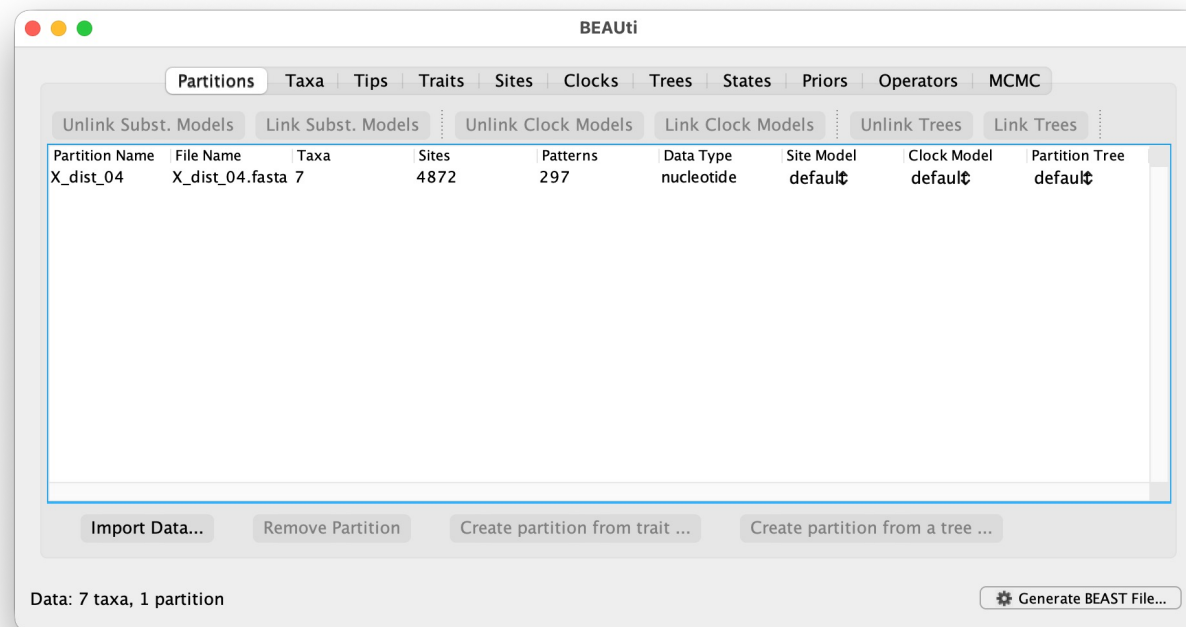
- What ML gave us (IQ-TREE, previous segment):
  - One tree per locus
  - Bootstrap support values
  - Discordance ACROSS loci
- What Bayesian gives us (BEAST X):
  - A posterior distribution of trees per locus
  - Captures uncertainty WITHIN a locus
  - Trees weighted by posterior probability
  - We can summarize as one tree OR visualize the full distribution

# We'll run BEAST X on a single locus: X\_dist\_04

- **Locus:** chrX:9,845,592 - 9,849,716 (4,872 bp), X-distal Xag region
- **Expected topology:** (arabiensis, quadriannulatus) sister, with christyi outgroup
- **From IQ-TREE earlier:** HKY+F+G4 best-fit model
- **Why this locus:** clean signal, fast Bayesian convergence, represents the “true species tree” topology
- **Why one locus only:** a Bayesian run on all 15 loci would take the full tutorial; we use one as a worked example

# BEAUi: import data and set substitution model

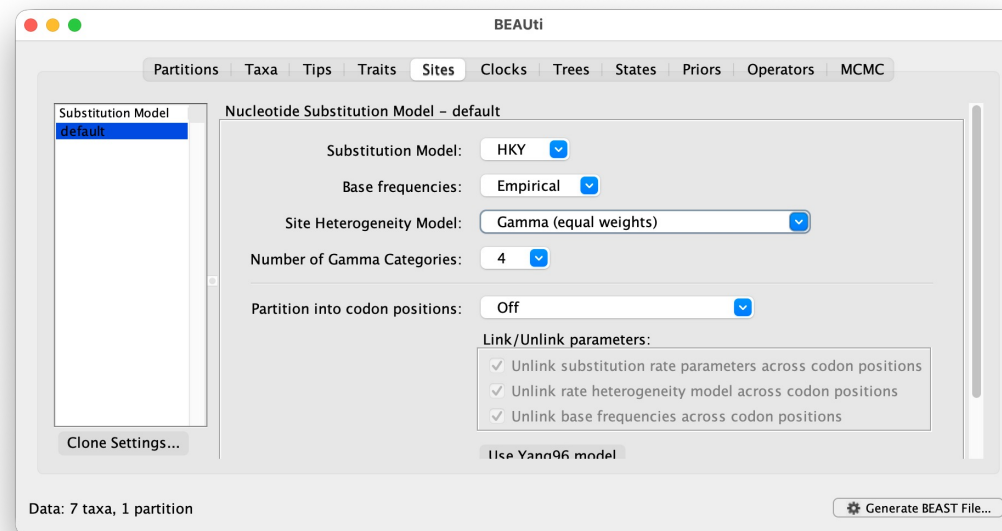
1. Launch BEAUi (from BEAST X installation)
2. File -> Import Data -> data/alignments/X\_dist\_04.fasta



# BEAUti: import data and set substitution model

## 3. Sites tab:

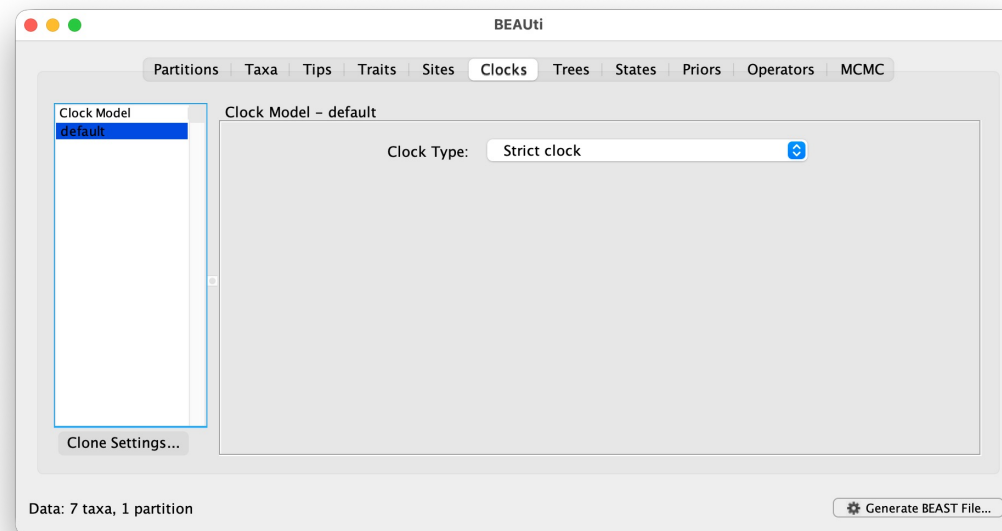
- Substitution Model: **HKY**
- Base frequencies: **Empirical**
- Site Heterogeneity Model: **Gamma**
- Number of Gamma Categories: **4**
- Partition into codon positions: **off**



# BEAUti: clock model and tree prior

## 4. Clocks tab:

- Clock Type: Strict clock

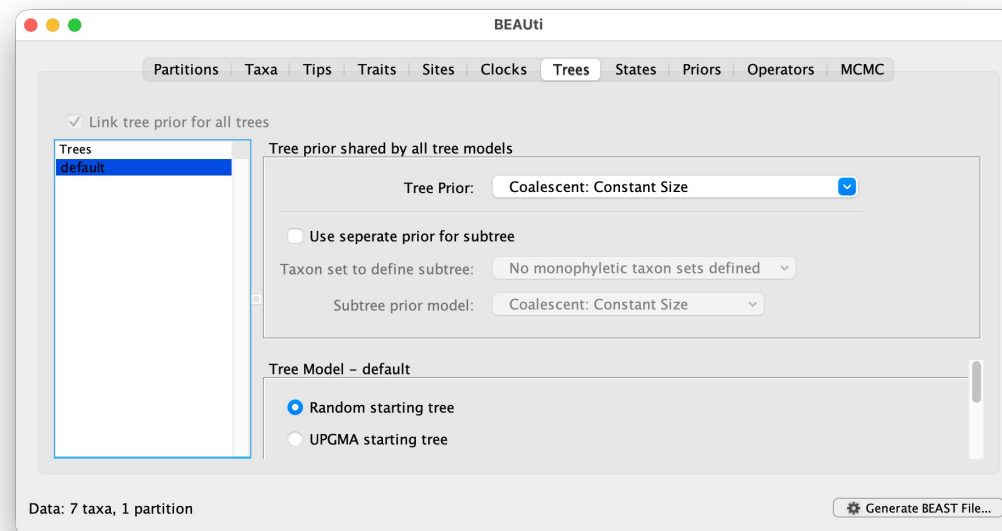




# BEAUti: clock model and tree prior

## 5. Trees tab:

- Tree Prior: **Coalescent: Constant Size**
- This is appropriate for the within-genus timescale



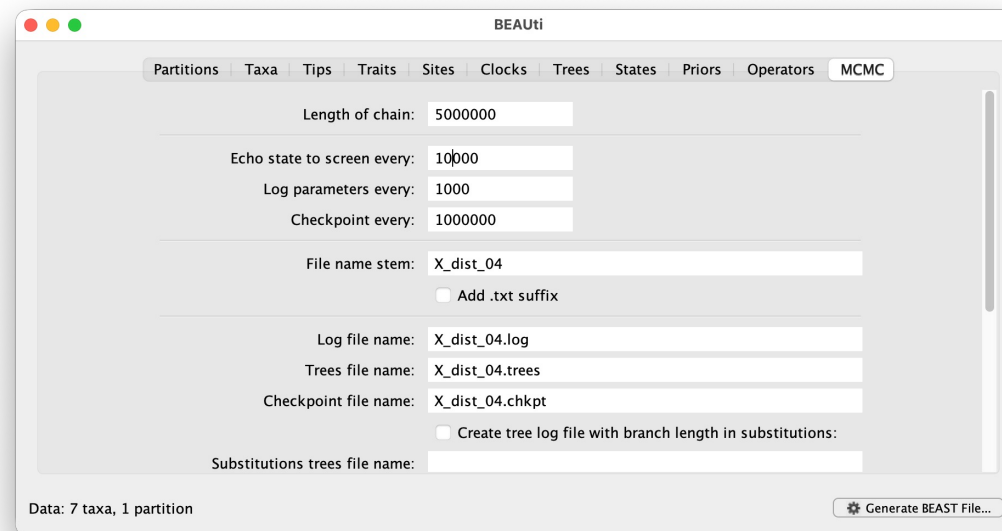
# BEAUti: clock model and tree prior

6. States tab: nothing to change
7. Priors tab: leave at defaults
8. Operators tab: leave at defaults

# MCMC settings, generate XML, and start BEAST

## 9. MCMC tab:

- Length of chain: **5,000,000** (short for live demo; production runs would use 20M)
- Echo state to screen every: 10,000
- Log parameters every: **1000** (gives 5,000 samples)
- File name stem: **X\_dist\_04**



# MCMC settings, generate XML, and start BEAST

10. File -> Generate XML -> save as `X_dist_04.xml`

11. Run BEAST X immediately in a terminal:

```
beast -beagle -overwrite X_dist_04.xml
```

- Expected runtime: about 2-3 minutes on Apple Silicon with BEAGLE
- Let it run in the background. We will come back to it later.

# Inspect a precomputed run in Tracer (while yours runs)

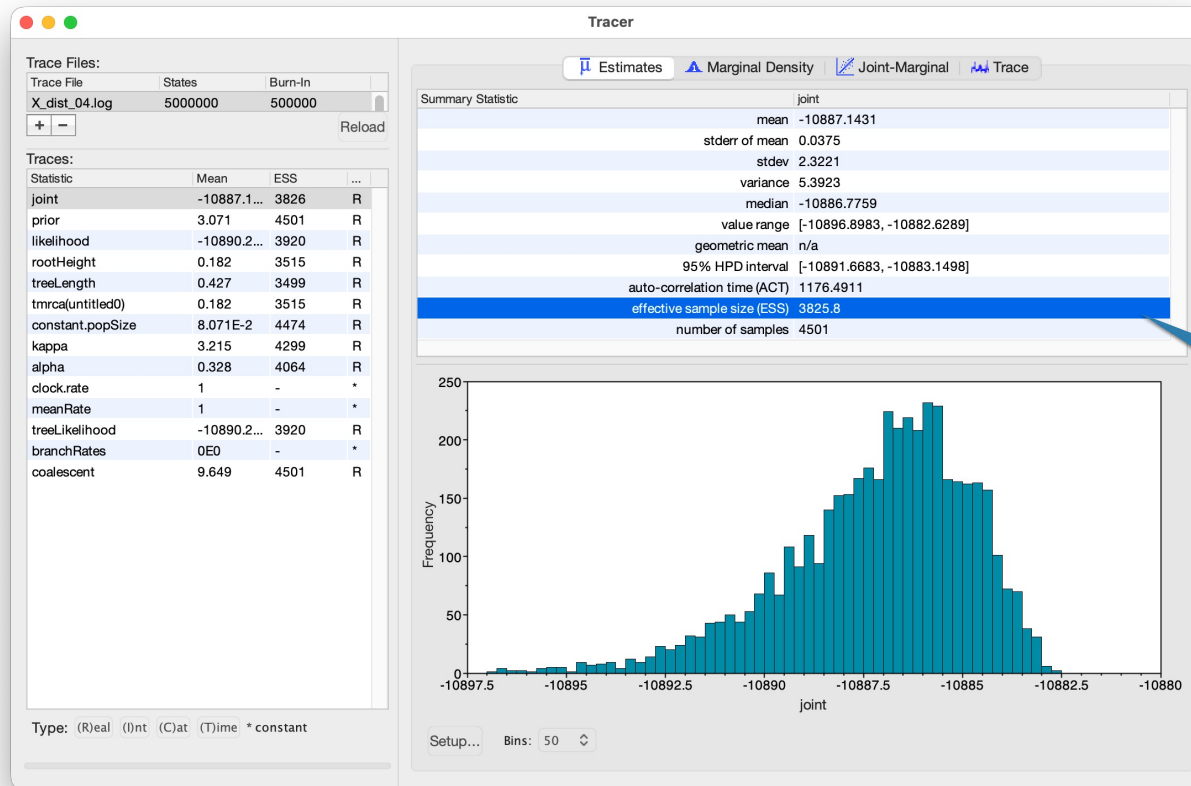
- Your live run is producing `X_dist_04.log` (parameter trace) and `X_dist_04.trees` (sampled trees), but it is not done yet
- To see what good convergence looks like NOW, open the precomputed long run from the repo:

Tracer

File -> Import Trace File

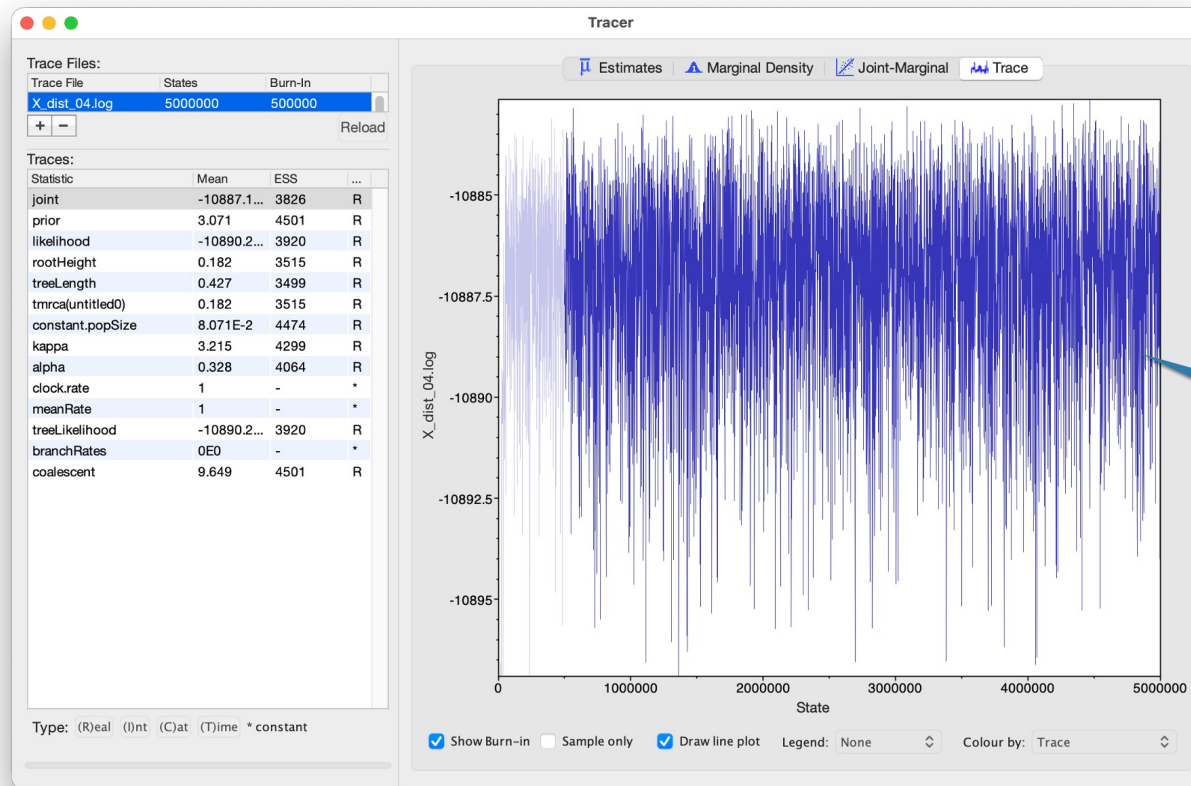
-> `precomputed/beast/X_dist_04.log`

# Inspect a precomputed run in Tracer (while yours runs)



Effective sample size is good (nearly 4000)

# Inspect a precomputed run in Tracer (while yours runs)



Sampling looks good  
("hairy caterpillar")

# Summarize the posterior with TreeAnnotator

- We have a distribution of trees but often want one summary tree
- TreeAnnotator computes the **Maximum Clade Credibility (MCC) tree**: the tree from the posterior whose clades collectively have the highest product of posterior support

```
treeannotator -burnin 500 -heights median \
precomputed/beast/X_dist_04.trees X_dist_04.MCC.tre
```

|                                   |                                                                       |
|-----------------------------------|-----------------------------------------------------------------------|
| -burnin 500                       | Discard first 500 trees                                               |
| -heights median                   | use median node heights for branch lengths                            |
| precomputed/beast/X_dist_04.trees | Input trees                                                           |
| X_dist_04.MCC.tre                 | one tree with posterior probabilities annotated on each internal node |

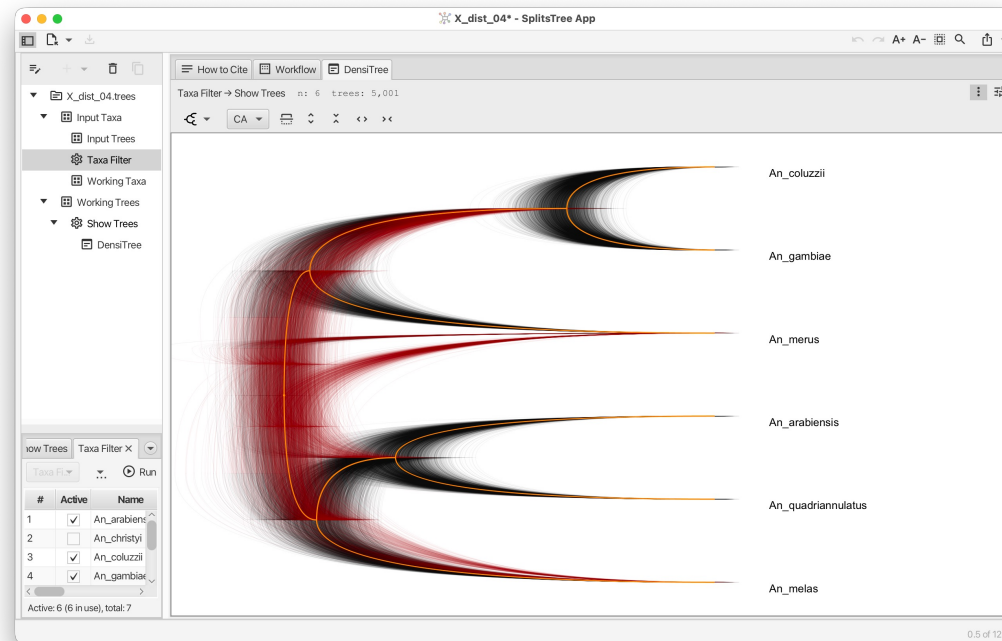


# Summarize the posterior with TreeAnnotator

- By the time you see this slide, your live BEAST run should be done.
- **Replace** `precomputed/beast/X_dist_04.trees` with `X_dist_04.trees` to use your own run instead.

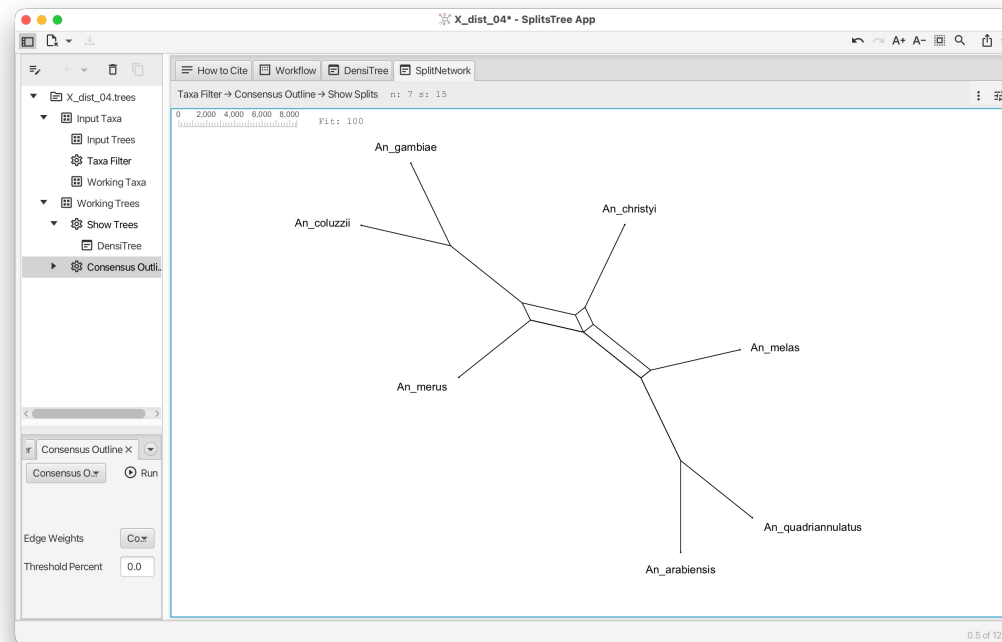
# Visualizing the full posterior: DensiTree in SplitsTree

- Open SplitsTree
- File -> Open -> select `X_dist_04.trees`  
(the full posterior, 5000 trees)
- Tree -> Show DensiTree to overlay all trees



# Your first network: consensus outline from the posterior

- In SplitsTree, with the same posterior tree set loaded:
  - Network menu -> Consensus Outline
- Network with splits weighted by posterior support (counts)





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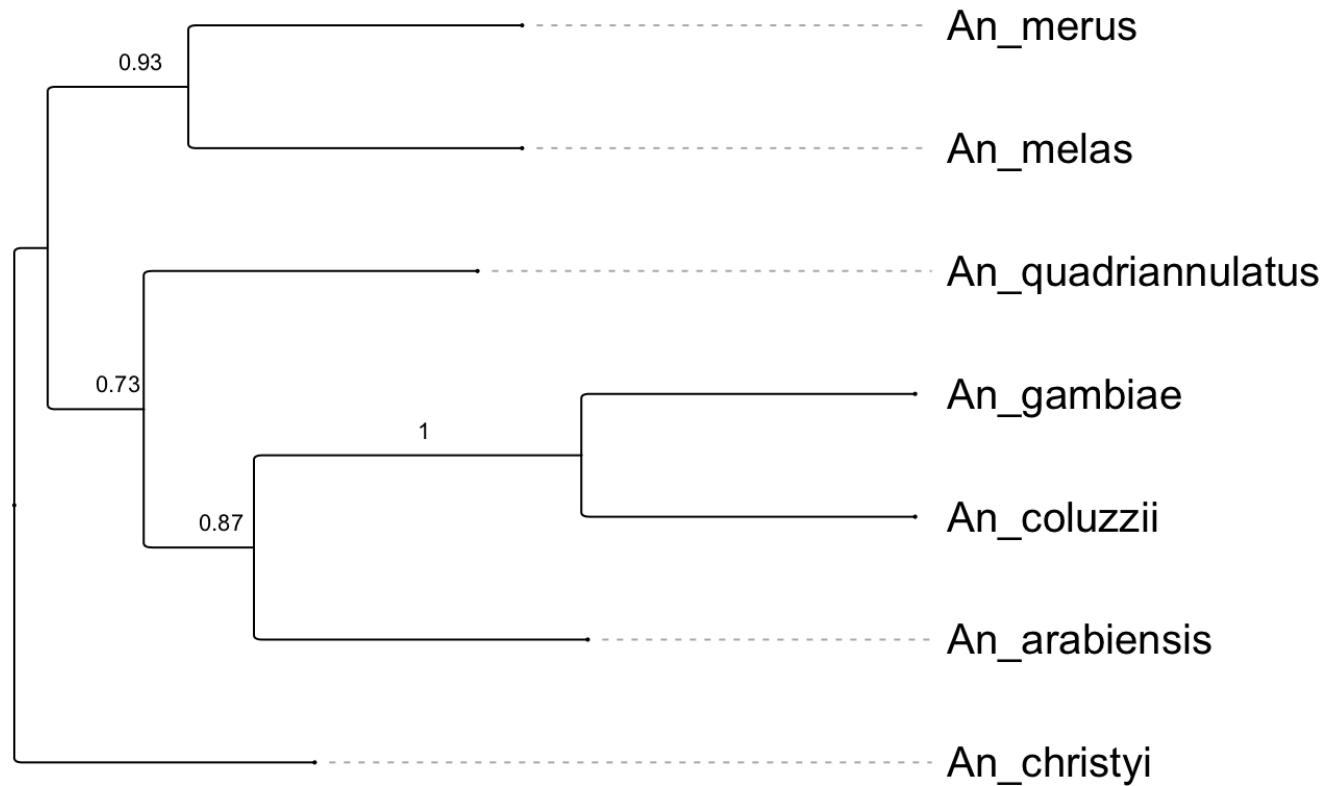
# Hands-on: coalescent species tree with ASTRAL

- **ASTRAL**: for a set of gene trees, returns a single species tree that maximizes quartet agreement under the multispecies coalescent (it models ILS but NOT introgression)
- **Input**: `tutorial_loci.treefile` (output of `iqtree`)
- **Command**:

```
java -jar astral.5.7.8.jar \  
    -i tutorial_loci.treefile \  
    -o tutorial_loci.ASTRAL_species_tree.tre
```

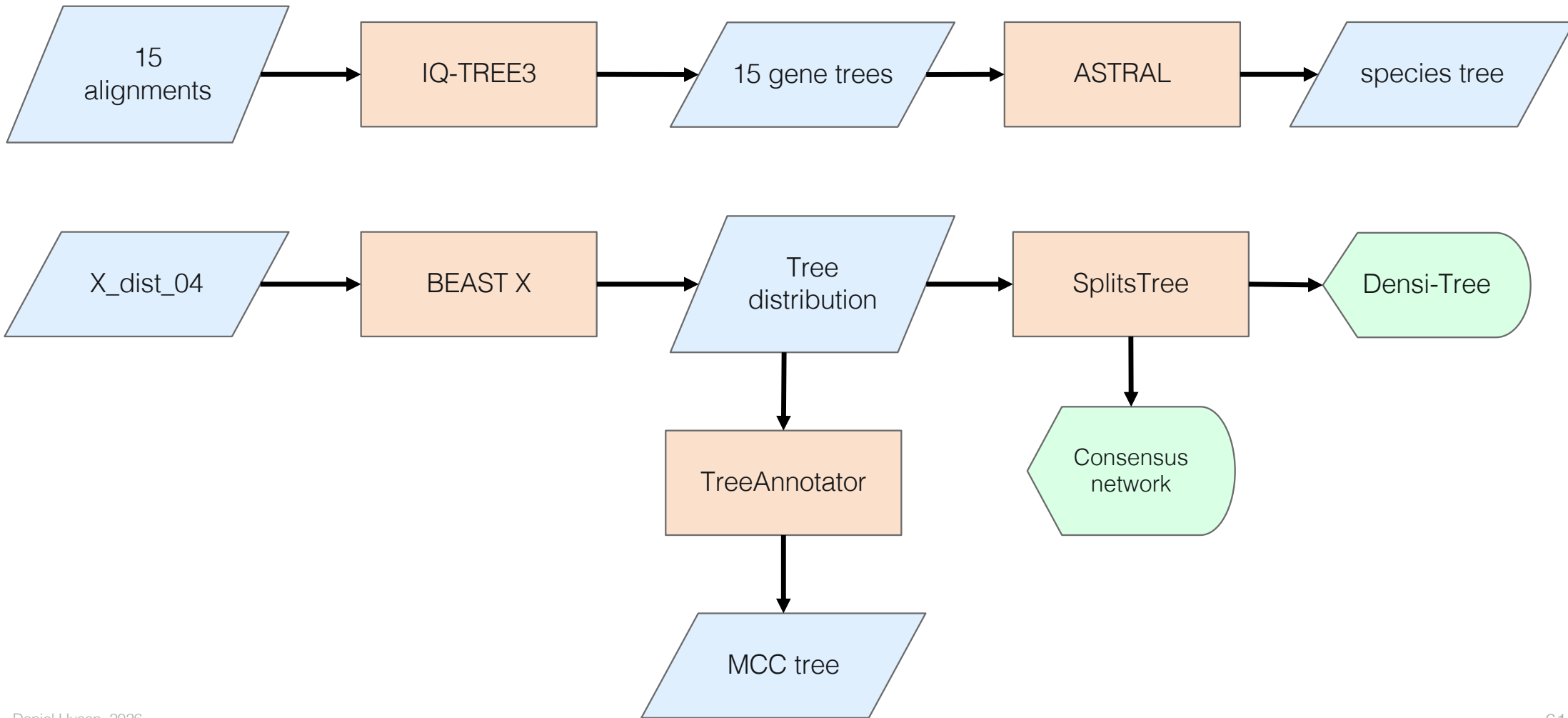
- Species tree with quartet support values at each internal node

# What does ASTRAL tell us?



(Re-rooted on *An\_christyi*; ASTRAL output is unrooted, so the Newick file looks different but the topology is the same.)

# Part I recap: from sequences to trees



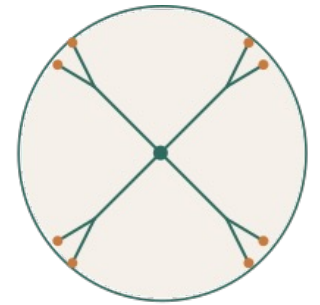
## But we know one of our gene tree categories is right

- The X-distal gene trees give the  
`(arabiensis, quadriannulatus)` topology
- We know from Fontaine that THIS is the true species branching order
- The autosomal gene trees and ASTRAL show the introgression-driven topology
- ASTRAL gave us a confident answer that is probably biologically wrong
- ASTRAL is *not broken* - it is doing exactly what its model says it should do
- The data contain *introgression* that the model (MSC, ILS only) can't represent



# PhyloGuide Q3

**Q:** When is a tree inadequate? When do we need a network?



**PhyloGuide**  
PHYLOGENETICS, GROUNDED

**A:** A tree is inadequate when evolution is not strictly branching, such as with hybridization, introgression, HGT, or strong conflicting signal. Networks capture mixed ancestry and conflicting histories.

# After the break: Making conflict explicit

- Part II preview:
  - SplitsTree Neighbor-Net: networks of conflicting splits in the alignment data
  - PhyloSketch: build a hypothesis network interactively
  - PhyloCompare: networks inferred from the gene trees - and SEE the introgression
- **Break:** 20 minutes, then 11:10 reconvene



COFFEE BREAK



# Share Your Feedback

Scan the QR code to let us  
know your thoughts on this  
tutorial.  
Thank you!



20 minutes





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10. PhyloCompare hands-on
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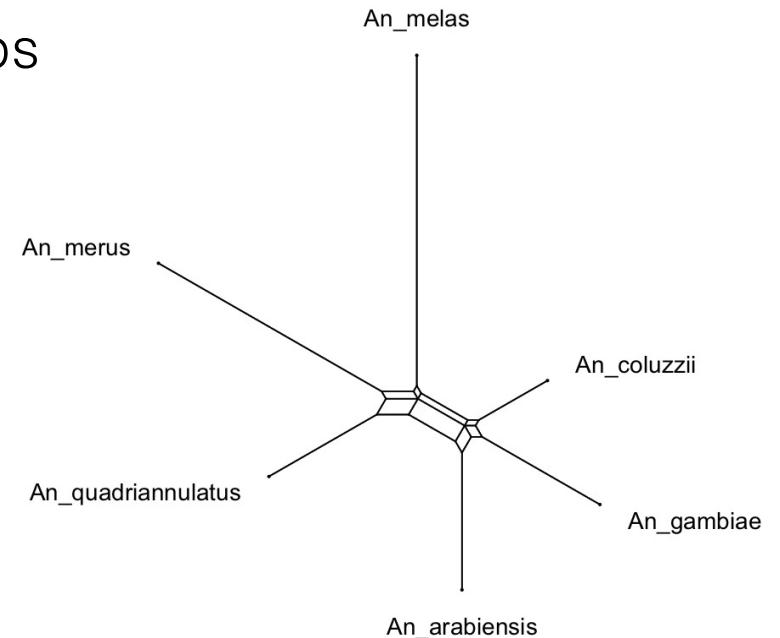


## Part II: Networks and cross-locus reticulation

- Where we left off:
  - 15 gene trees, systematically disagreeing
  - ASTRAL gave a confident species tree
  - But it is the introgression-driven topology, not the true one
  - We have evidence the data contain a non-tree-like signal
- What we will do next:
  - Three network views of our data, each with a different tool
  - Networks visualize the conflict explicitly
  - We will *see* the introgression rather than infer it indirectly

# Split networks

- A **split** divides the set of taxa into two groups
  - Each edge in a phylogenetic tree corresponds to a split
- A tree has a **compatible** set of splits
  - no two splits are incompatible
- A split network can represent **incompatible** splits
  - Any two incompatible splits produce a parallelogram in a split network



# Building a split network from a distance matrix

- Input: a pairwise distance matrix (computed from an alignment)
- Neighbor-Net (Bryant & Moulton 2004) extends Neighbor-Joining
- Instead of forcing each step to produce a tree, it can retain conflicting evidence as parallelograms
- Output: a split network that approximates the input distances
- Tool: **SplitsTree** (you already have it open from Part I)

# Two roles for networks: implicit and explicit

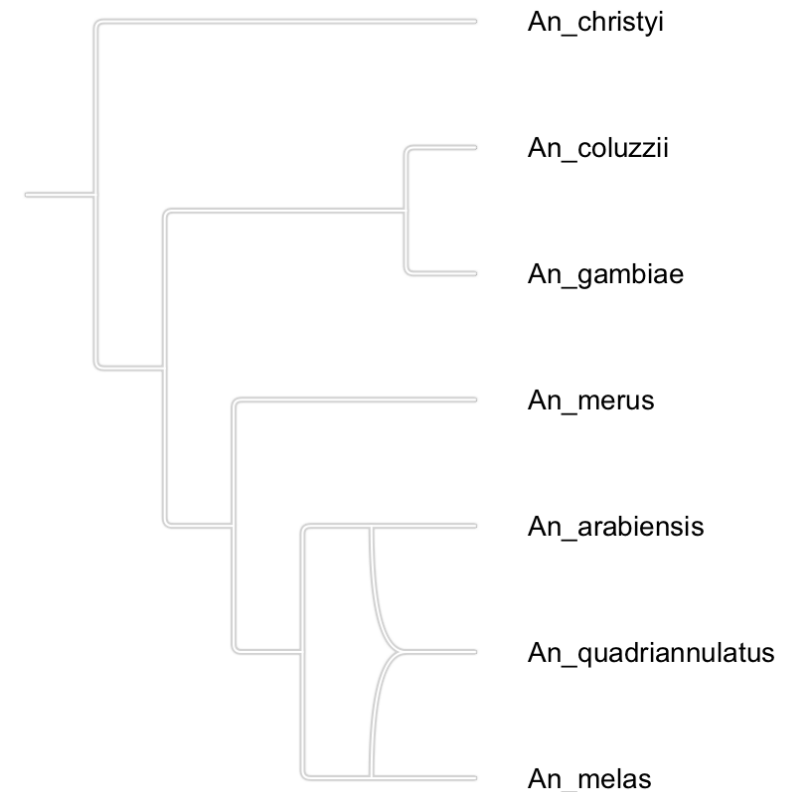
## 1. Implicit networks

- Visualize conflicting signal without claiming specific evolutionary events
- “Here is conflict” - they don’t say what caused it
- Examples: split networks
- Exploration, visualization, detecting non-tree-like signal



# Rooted phylogenetic networks

- A rooted **phylogenetic tree** is a tree with
  - One root, leaves are taxa, internal nodes are inferred ancestors
- A **phylogenetic network** generalizes this:
  - Internal nodes can have more than one parent (**reticulation nodes**)
  - Can represent both vertical inheritance AND events where lineages combine or exchange genetic material.



# Two roles for networks: implicit and explicit

## 2. Explicit networks

- Represent specific evolutionary events: hybridization, introgression, horizontal transfer
  - Have reticulation nodes corresponding to those events
  - “These two parent lineages produced this hybrid”
  - Examples: PhyloFusion output, hybridization networks
- Inferring biological history

# Explicit reticulation: where two lineages meet

- A **reticulation node** in an explicit network represents an event where genetic material from two parent lineages combined into one descendant
- Biologically, this may correspond to:
  - **Hybridization**: two species produce a hybrid that becomes a new lineage
  - **Introgression**: gene flow between two species after their initial divergence
  - **Horizontal gene transfer**: genetic material moves between non-sister taxa
  - **Recombination**: blocks of DNA from different parents
- A network can have multiple reticulation nodes - one per event

# You have already computed a network

- In Part I, the consensus network from the BEAST X posterior:
  - Each split weighted by posterior probability
  - Parallelograms appear where the posterior is uncertain
  - This was an **implicit** network of within-locus uncertainty
- Today we will compute several more, all looking at our 15 gene trees from different angles



# From gene trees to a reticulate network

- PhyloFusion (Zhang, Cetinkaya, Huson 2026) takes as input a set of rooted gene trees and computes an **explicit network**
- Finds a network that displays all input trees with as few reticulations as possible
- Each reticulation in the output corresponds to a hypothesized reticulation event that explains why the gene trees disagree
- Interactive: you can select which gene trees to include in the inference, and which to display *inside* the inferred network

# Our three network views today

1. **SplitsTree Neighbor-Net on alignments** (next, 20 min)
  - Input: alignment data
  - Output: implicit split network of conflicting splits
  - Goal: see the raw conflict signal in the data
2. **PhyloSketch interactive sketching** (30 min in)
  - Input: your biological intuition + the gene trees you have seen
  - Output: a hand-drawn hypothesis network
  - Goal: think like a phylogeneticist building a hypothesis
3. **PhyloCompare on the gene tree set** (60 min in)
  - Input: the 15 gene trees from IQ-TREE
  - Output: explicit reticulation network (using PhyloFusion)
  - Goal: see introgression events inferred directly from the trees



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# Hands-on: Neighbor-Net on the 15 alignments

- Time budget: 20 minutes
- Goal: visualize the conflicting signal directly in alignment data, before any gene tree is computed using **SplitsTree**
- Plan:
  1. Walk through Neighbor-Net on one alignment together
  2. Each participant loads 2-3 more alignments from different categories and compares
  3. Discuss what the network shapes reveal



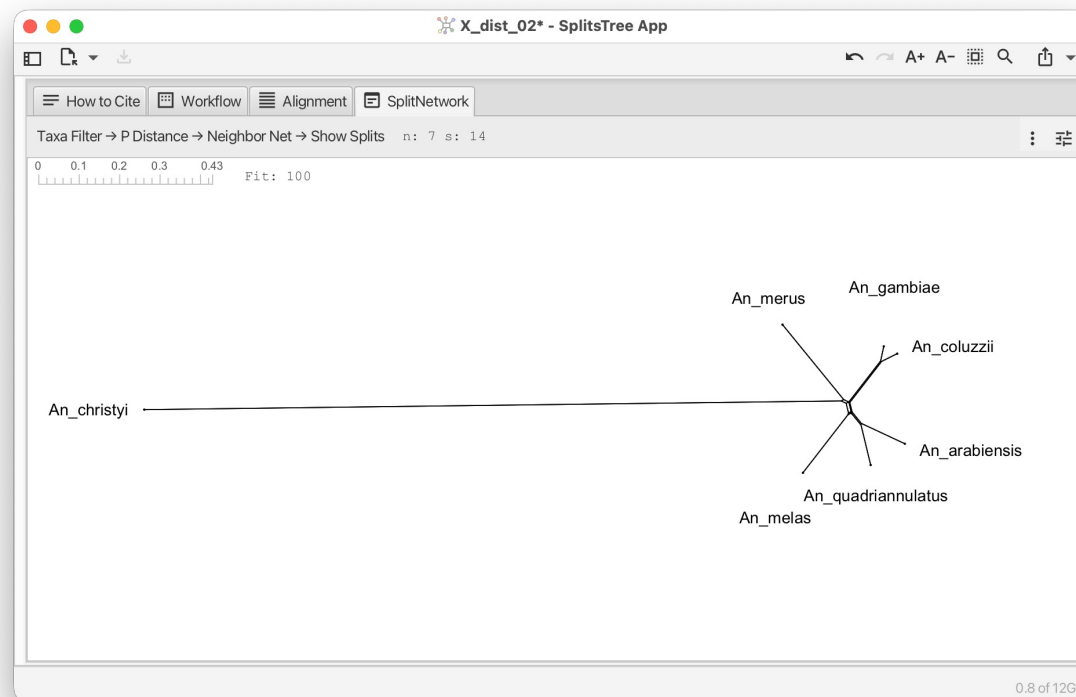
# Neighbor-Net on our alignments

- We can use SplitsTree+ Neighbor-Net to compute split networks for our 15 alignments
- Question: do they all look the same? Where are the parallelograms?

| Region                      | Loci | Expected topology                                      |
|-----------------------------|------|--------------------------------------------------------|
| X distal<br>(Xag inversion) | 5    | (arabensis, quadriannulatus) -<br>species tree         |
| X pericentromeric           | 1    | (arabensis, (gambiae,<br>coluzzii)) - introgressed     |
| Autosomal (2R, 3R)          | 5    | (arabensis, (gambiae,<br>coluzzii)) - introgressed     |
| 2La inversion (2L)          | 2    | 2La-specific topologies                                |
| 3La inversion (3L)          | 2    | (merus, quadriannulatus) -<br>unexpected introgression |

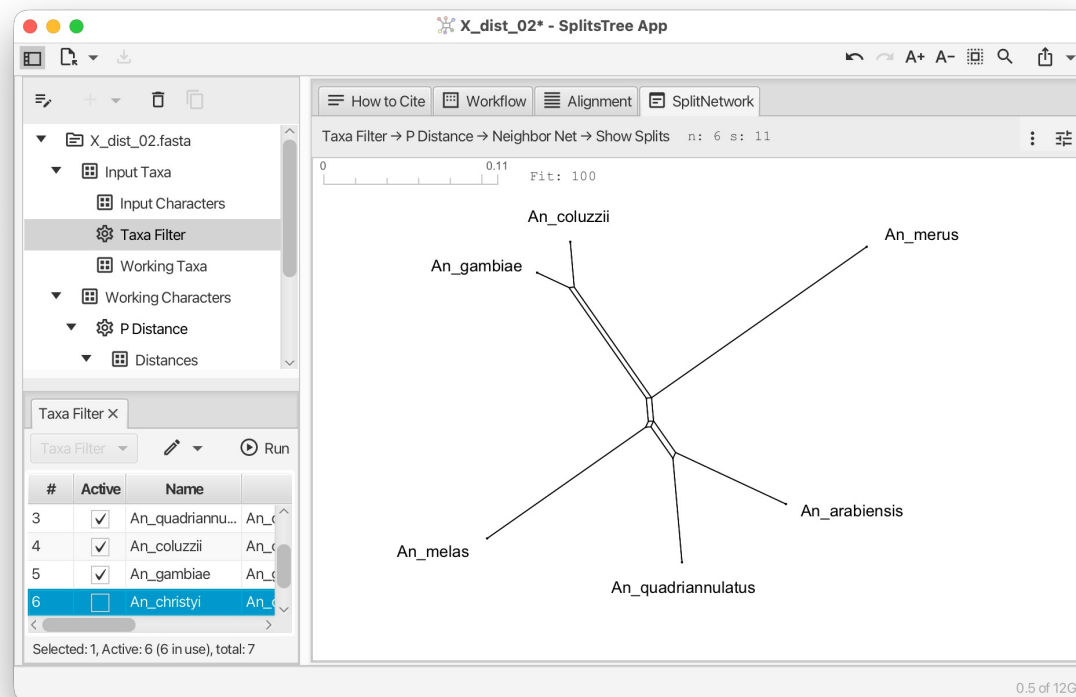
# Let's do one together: X\_dist\_02

- Open file `alignments/X_dist_02.fasta` in SplitsTree
- This will run p-distances and Neighbor-net



# Let's do one together: X\_dist\_02

- Open file `alignments/X_dist_02.fasta` in SplitsTree
- Use the Taxa Filter item (side bar) to deactivate the outgroup:

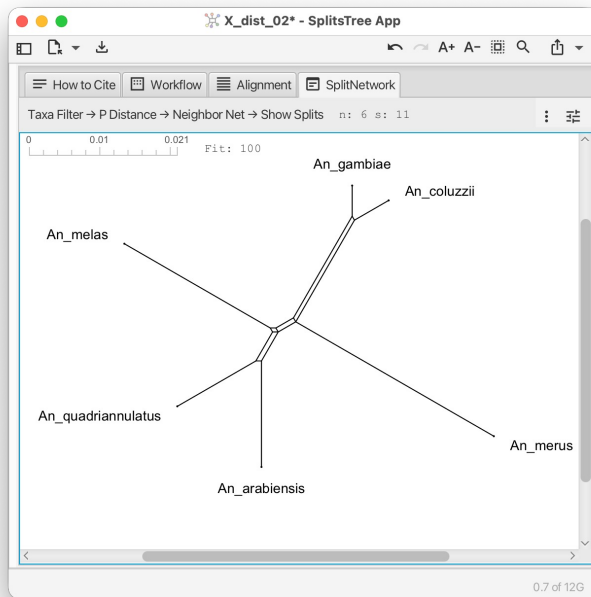


# Open other alignments and explore

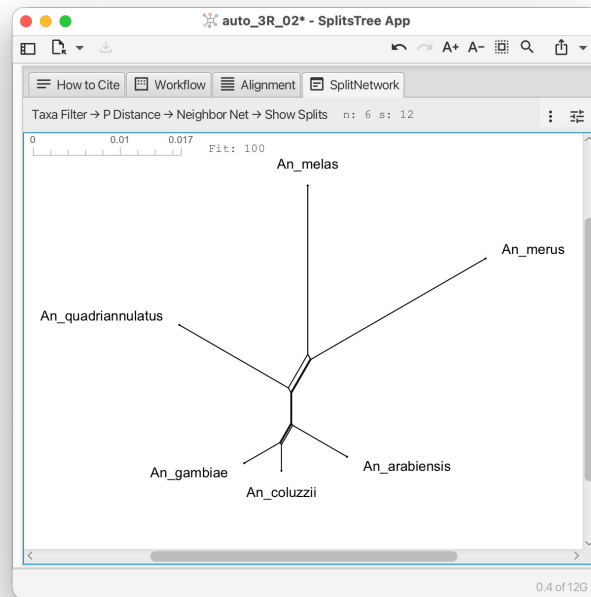
- Repeat the same workflow on alignments from different categories:

| Region                      | Loci | Expected topology                                      |
|-----------------------------|------|--------------------------------------------------------|
| X distal<br>(Xag inversion) | 5    | (arabensis, quadriannulatus) -<br>species tree         |
| X pericentromeric           | 1    | (arabensis, (gambiae,<br>coluzzii)) - introgressed     |
| Autosomal (2R, 3R)          | 5    | (arabensis, (gambiae,<br>coluzzii)) - introgressed     |
| 2La inversion (2L)          | 2    | 2La-specific topologies                                |
| 3La inversion (3L)          | 2    | (merus, quadriannulatus) -<br>unexpected introgression |

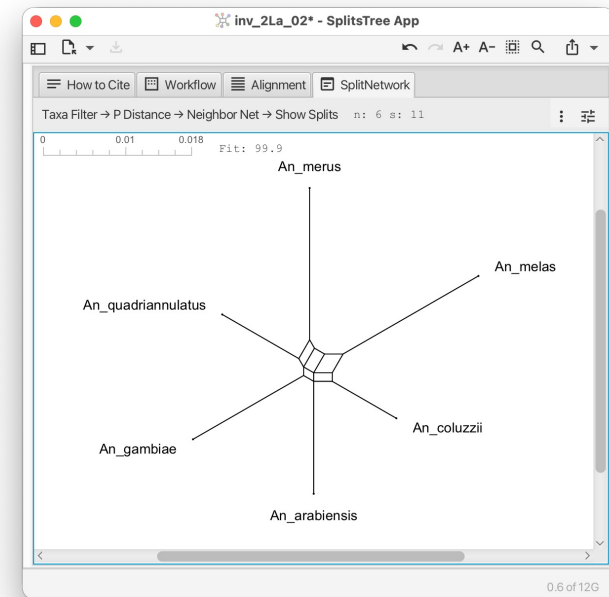
# Open other alignments and explore



- X\_dist\_02
- “species tree”

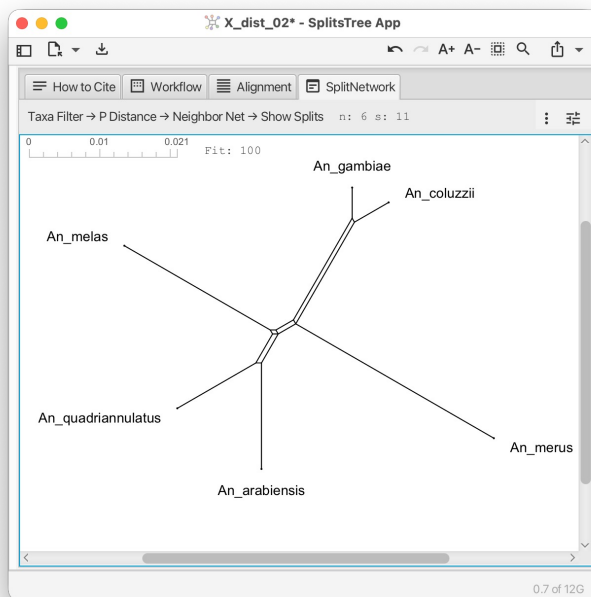


- auto\_3R\_02
- “introgressed”

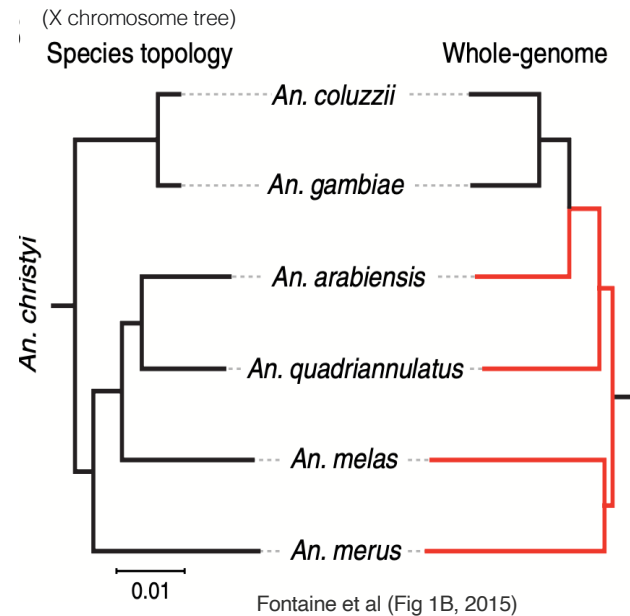


- Inv\_2La\_02
- “2La-specific topologies”

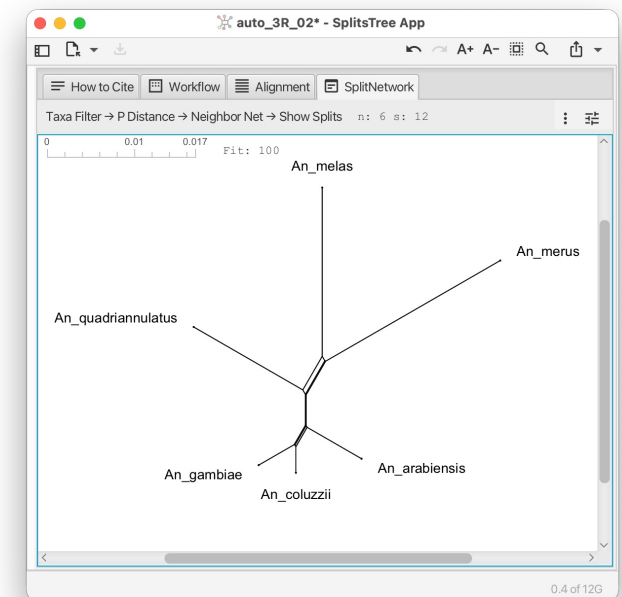
# Compare with trees in Fontaine et al (2015)



- X\_dist\_02
- “species tree”

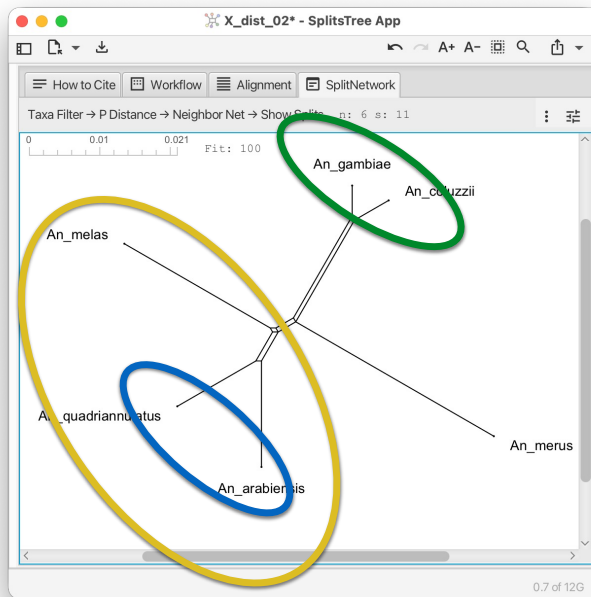


Fontaine et al (Fig 1B, 2015)

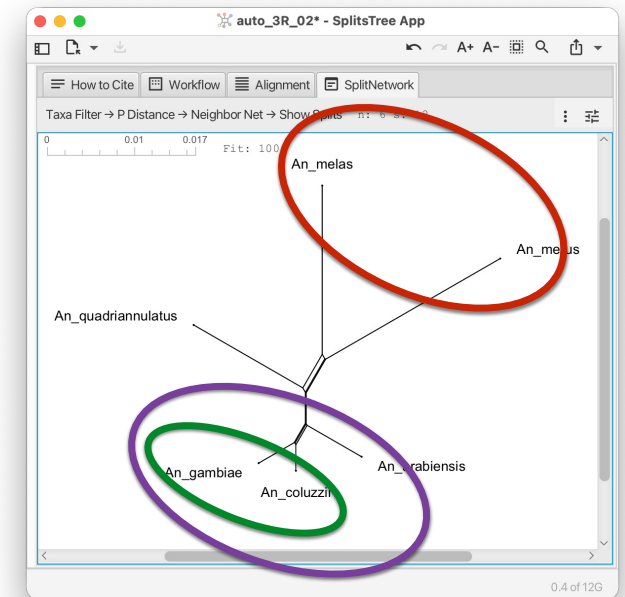
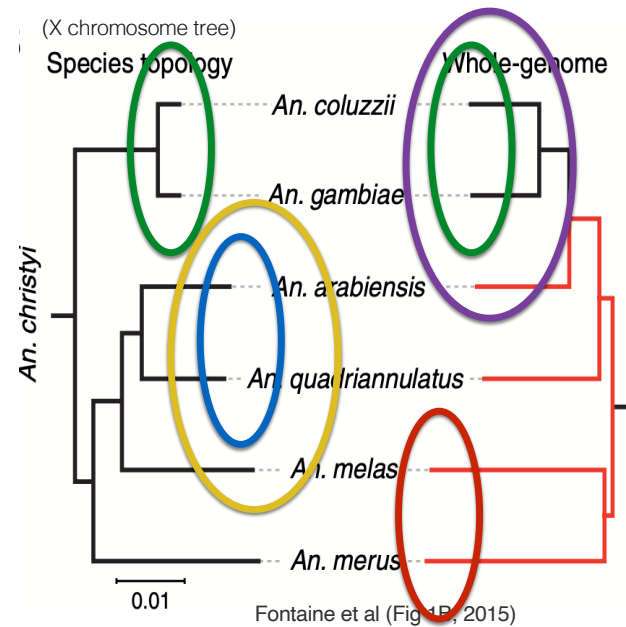


- auto\_3R\_02
- “introgressed”

# Compare with trees in Fontaine et al (2015)

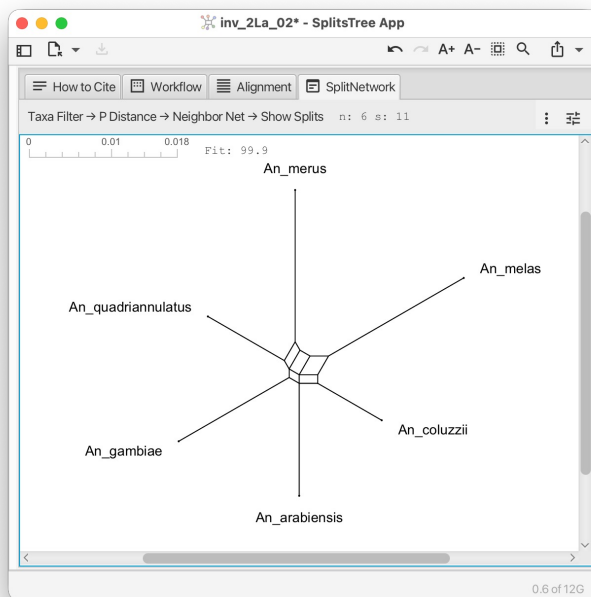


- X\_dist\_02
- “species tree”

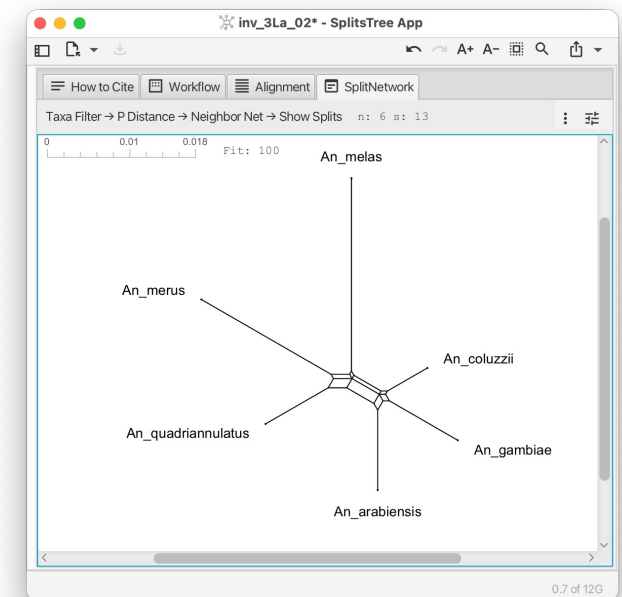
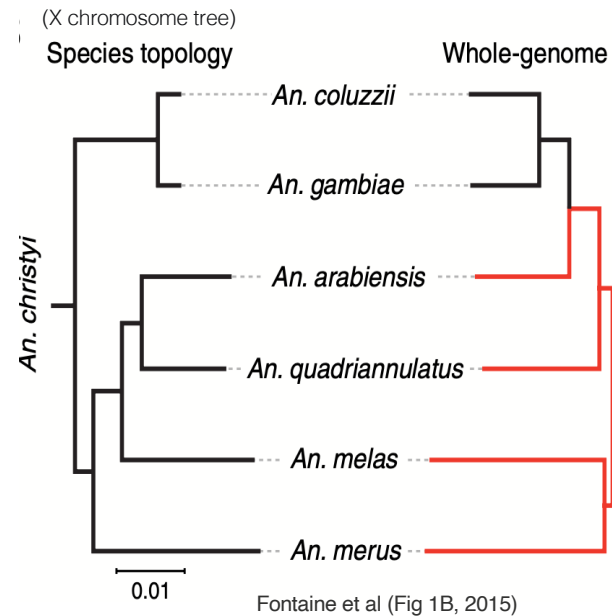


- auto\_3R\_02
- “introgressed”

# Compare with trees in Fontaine et al (2015)



- Inv\_2La\_02
- “2La-specific topologies”



- Inv\_3La\_02
- “unexpected introgression”



# What did you see?

- Across the locus categories, you should have seen:
  - **X-distal** alignments: smaller parallelograms; closer to tree-like
  - **Autosomal** alignments: more parallelograms involving `arabensis` and the (`gambiae`, `coluzzii`) clade
  - **2La inversion** alignments: distinctive boxes reflecting the trans-specific polymorphism
  - **3La inversion** alignments: parallelograms grouping `merus` with `quadriannulatus`
- The **big-picture observation**: the conflict pattern we saw in gene trees from IQ-TREE is also visible directly in alignment data



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# And what's next?

Bridge to PhyloSketch (next, 20 min):

- Neighbor-Net produces an **implicit** network - it indicates conflict
- For our next step, we will build an **explicit** network by hand, drawing reticulation nodes representing specific hypothesized evolutionary events
- This is the “phylogenetic sketching” workflow

# Hands-on: sketch a hypothesis network in PhyloSketch

- **Time budget:** 20 minutes
- **Goal:** build, by hand, a phylogenetic network that captures your best hypothesis about Anopheles evolution given everything we've seen
- **Tool:** PhyloSketch (Huson 2025, PLOS Comp Biol)
- **The pedagogical idea:**
  - You have observed **patterns** in gene trees and Neighbor-Nets
  - Translate those patterns into a **biological hypothesis**: what actually happened in this group's evolutionary history?
  - Draw your hypothesis as an explicit network with reticulation nodes

# PhyloSketch in 90 seconds

- **Launch PhyloSketch**
- **Drawing leaves and branches:** click to place taxa, drag to connect them
- **Building a tree:** connect leaves through internal nodes
- **Adding a reticulation:** convert an internal node to a reticulation node, then connect a *second* parent edge to it
- **Editing labels:** double-click to rename
- **Saving:** File -> Save As -> .psketch file

# Build your hypothesis network

**Step 1: Start from the species tree we believe is correct**

From X-distal loci, the species tree is:

```
((((arabiensis,quadriannulatus),
(gambiae, coluzzii)),melas),merus);
```

with *christyi* as outgroup. Sketch this in PhyloSketch first.

**Step 2: Add reticulation nodes for hypothesized introgression**

Events to consider:

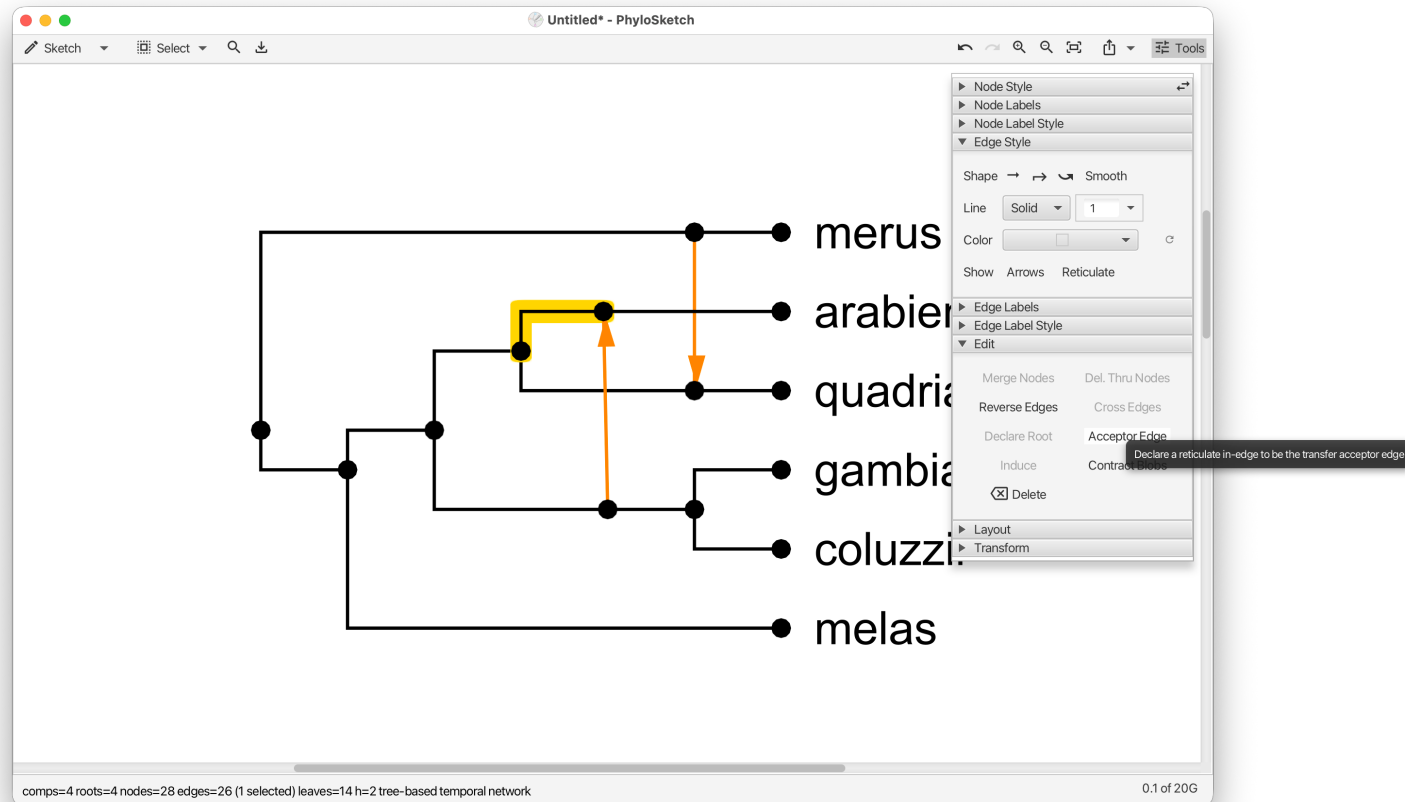
| Event                                                  | Evidence in our data                                                                             |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Introgression from (gambiae, coluzzii) into arabiensis | Autosomal gene trees show arabiensis grouped with (gambiae, coluzzii) instead of quadriannulatus |
| Introgression between merus and quadriannulatus        | 3La inversion gene trees show (merus, quadriannulatus) sister                                    |
| Bidirectional gene flow involving the 2La inversion    | 2La inversion gene trees show distinctive non-species-tree topologies                            |

# Build your hypothesis network

## Step 3: Decide which events to include

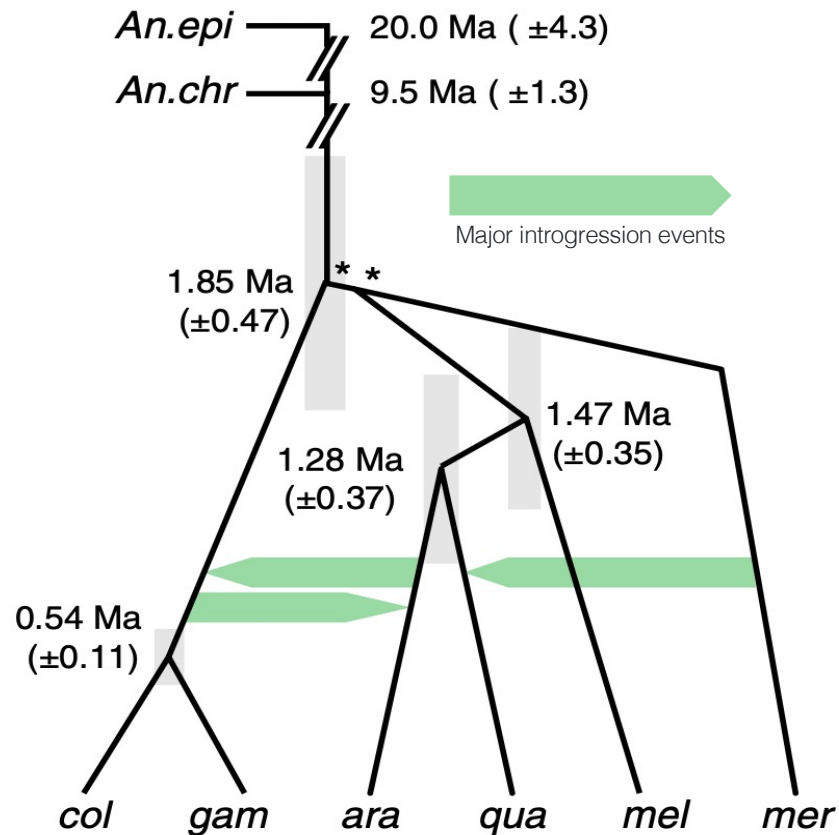
- A minimal network has 1 reticulation
- A richer network might have 2 or 3
- Trade-off: every reticulation must be justified by evidence

# Build your hypothesis network

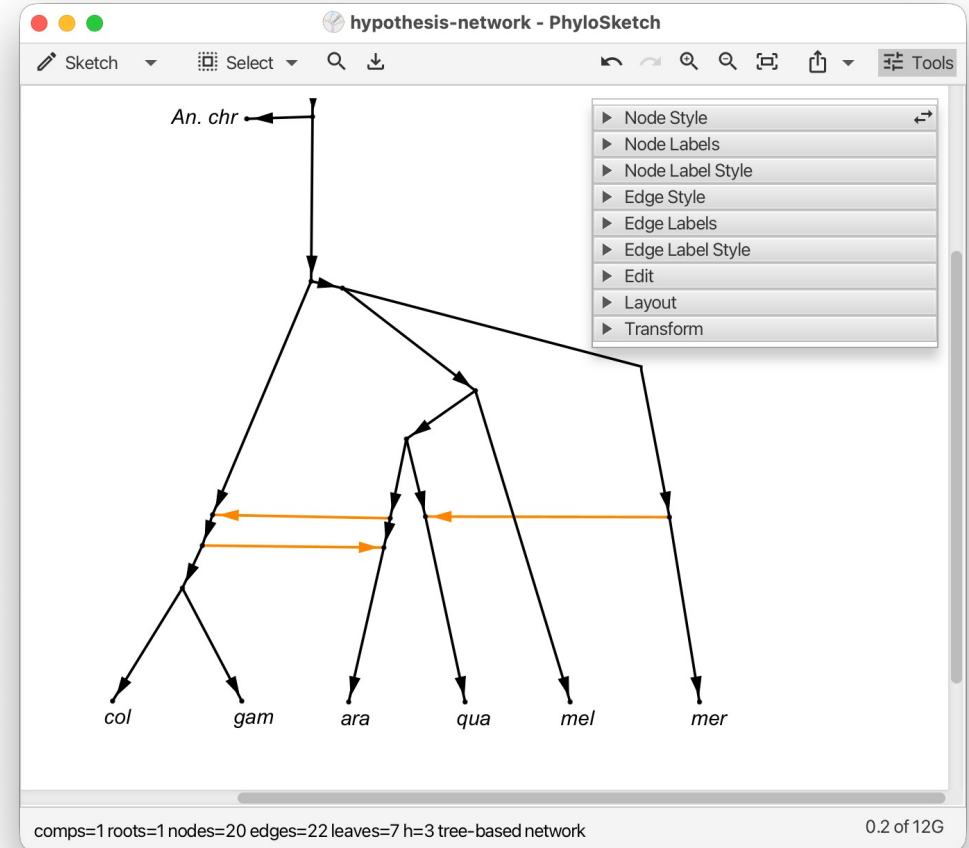




# Answer provided in paper (do not peek too early)



Fontaine et al (Fig 1C, 2015)





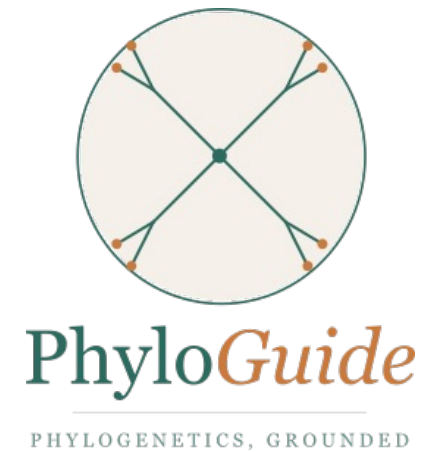
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# PhyloGuide Q4

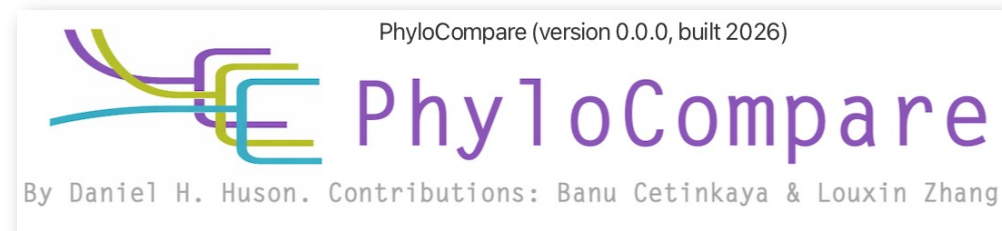
**Q:** We want to infer an explicit reticulation network from a set of gene trees, with the minimum number of reticulations needed to display all the trees. What method should we use?

**A:** Use a minimum-hybridization network method, which finds the smallest number of reticulation events explaining all gene trees. Check ILS first because it can mimic reticulation.



# Hands-on PhyloCompare: from gene trees to a network

- **Time budget:** 25 minutes
- **Goal:** compute an explicit reticulation network from the 15 gene trees you produced in Part I, then interactively explore how different subsets of trees give different networks
- **The big question for this segment:** does the network agree with what you sketched in PhyloSketch?



# Hands-on PhyloCompare : from gene trees to a network

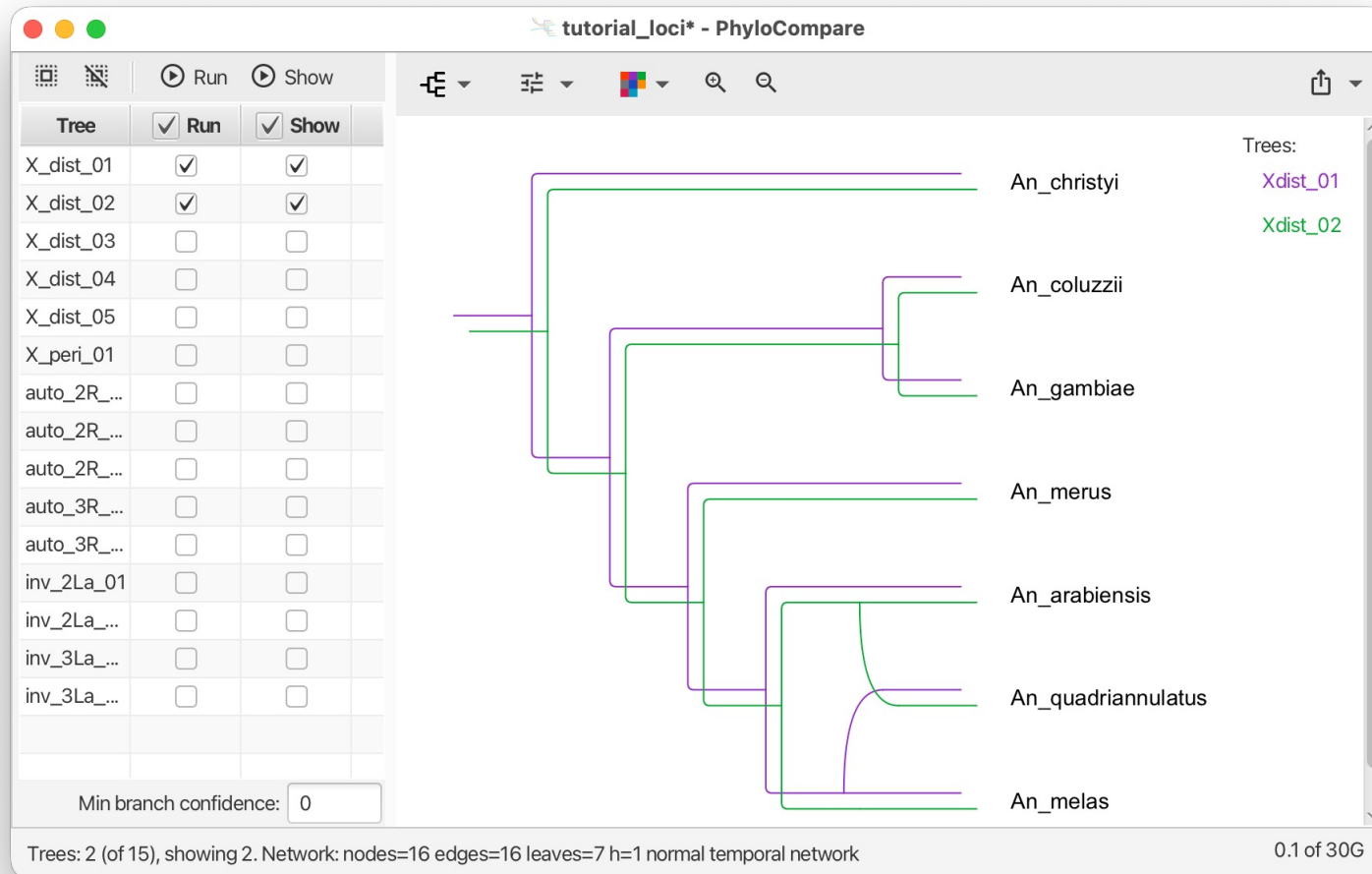
## PhyloCompare

- Novel comparison tool for trees (under development)
- It uses the PhyloFusion algorithm to compute a minimum-reticulate network that contains all input trees (Zhang et al (2026), *Systematic Biology*)
- We will use the tool for network inference
- Each reticulation in the output is a hypothesized event - matching a hybridization, introgression, or other gene-flow event

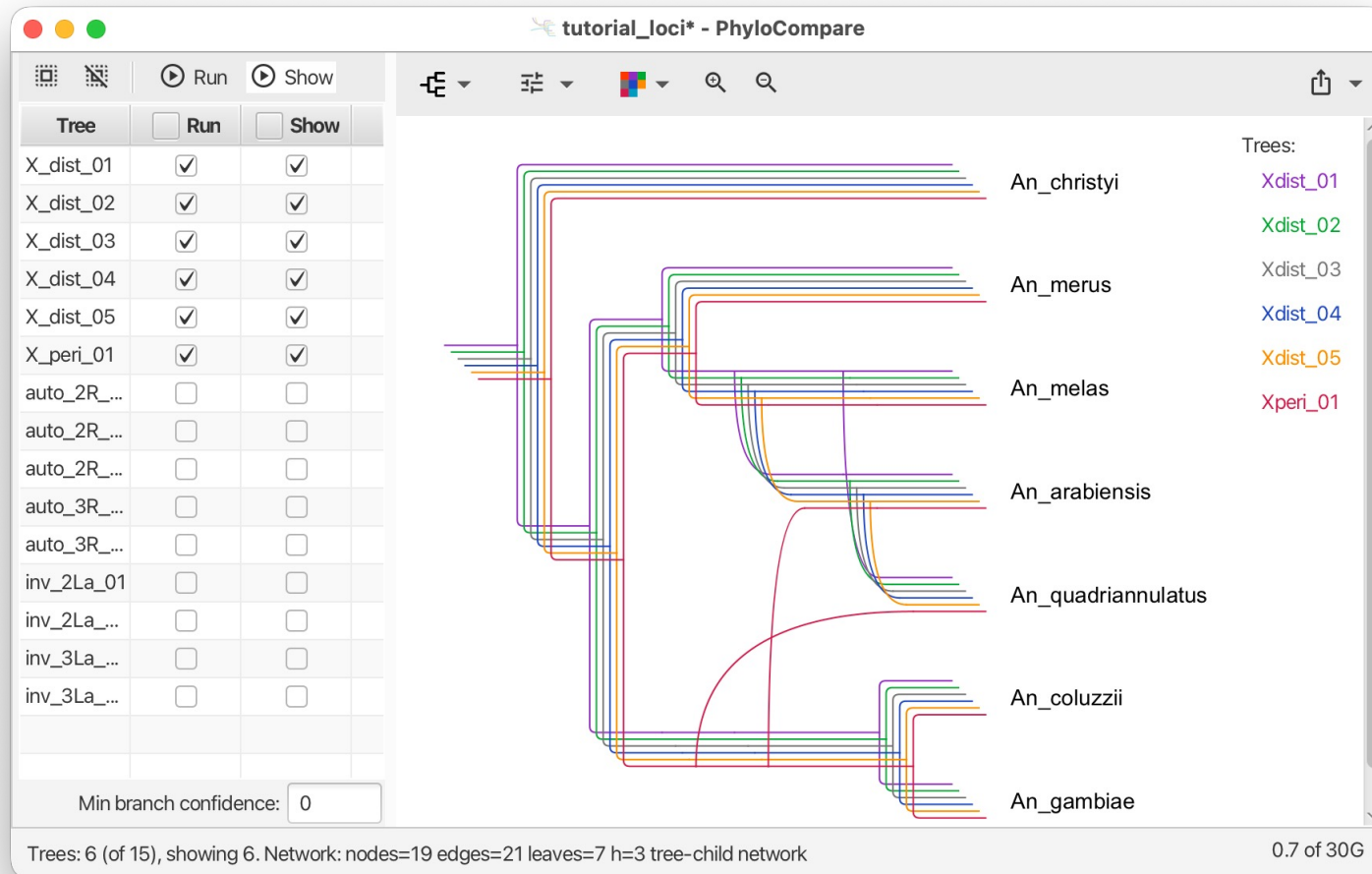
# Hands-on PhyloCompare: from gene trees to a network

1. Launch PhyloCompare
2. File -> Open -> select `tutorial_loci.nex` (the NEXUS tree set we created after IQ-TREE; it contains all 15 gene trees with their locus IDs)
3. PhyloCompare should list all 15 trees in the tree-set panel; by default, the first trees are combined into a network
4. Explore different sets of trees to run comparisons on

# Hands-on PhyloCompare : from gene trees to a network

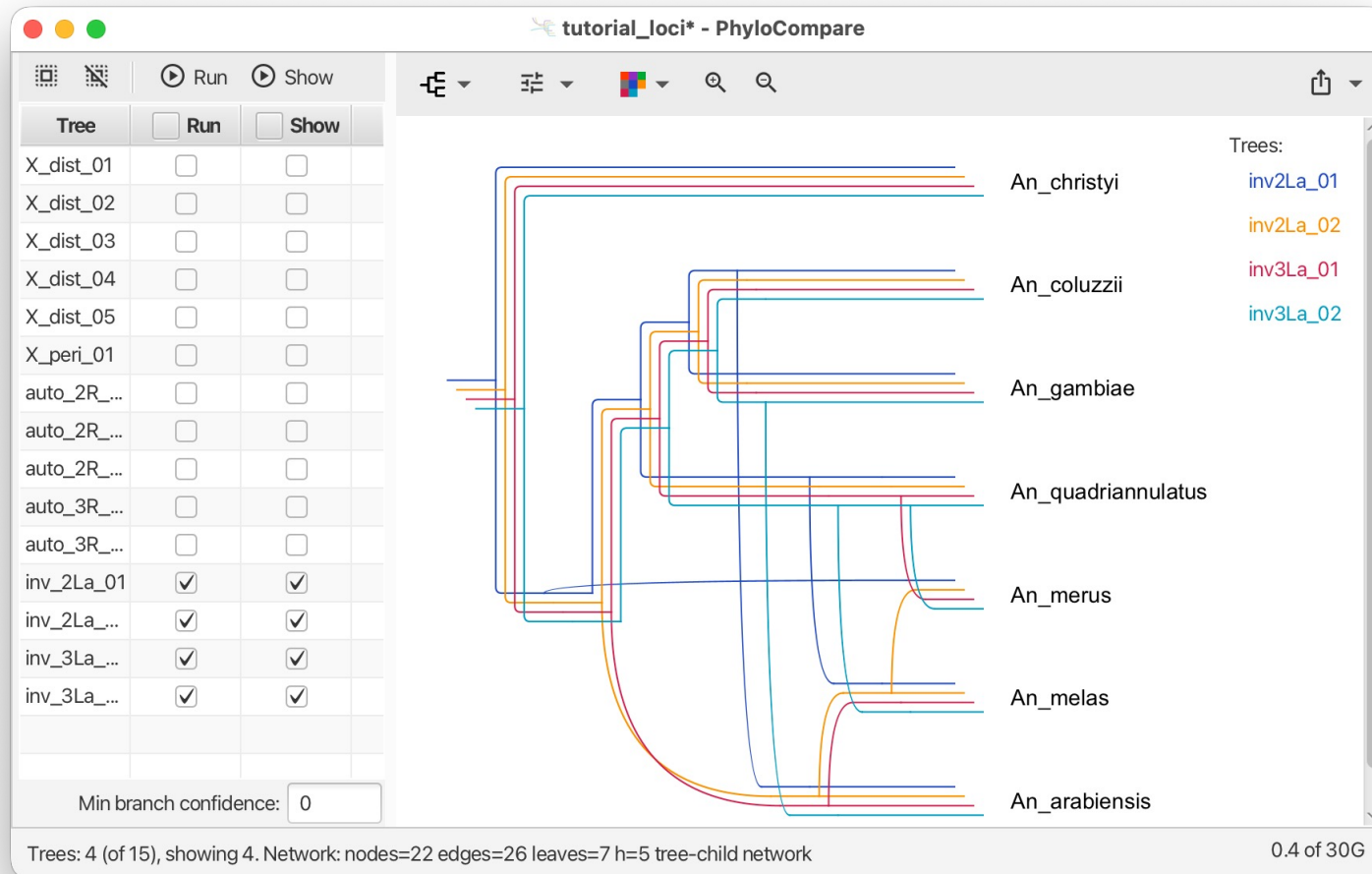


# Hands-on PhyloCompare : from gene trees to a network

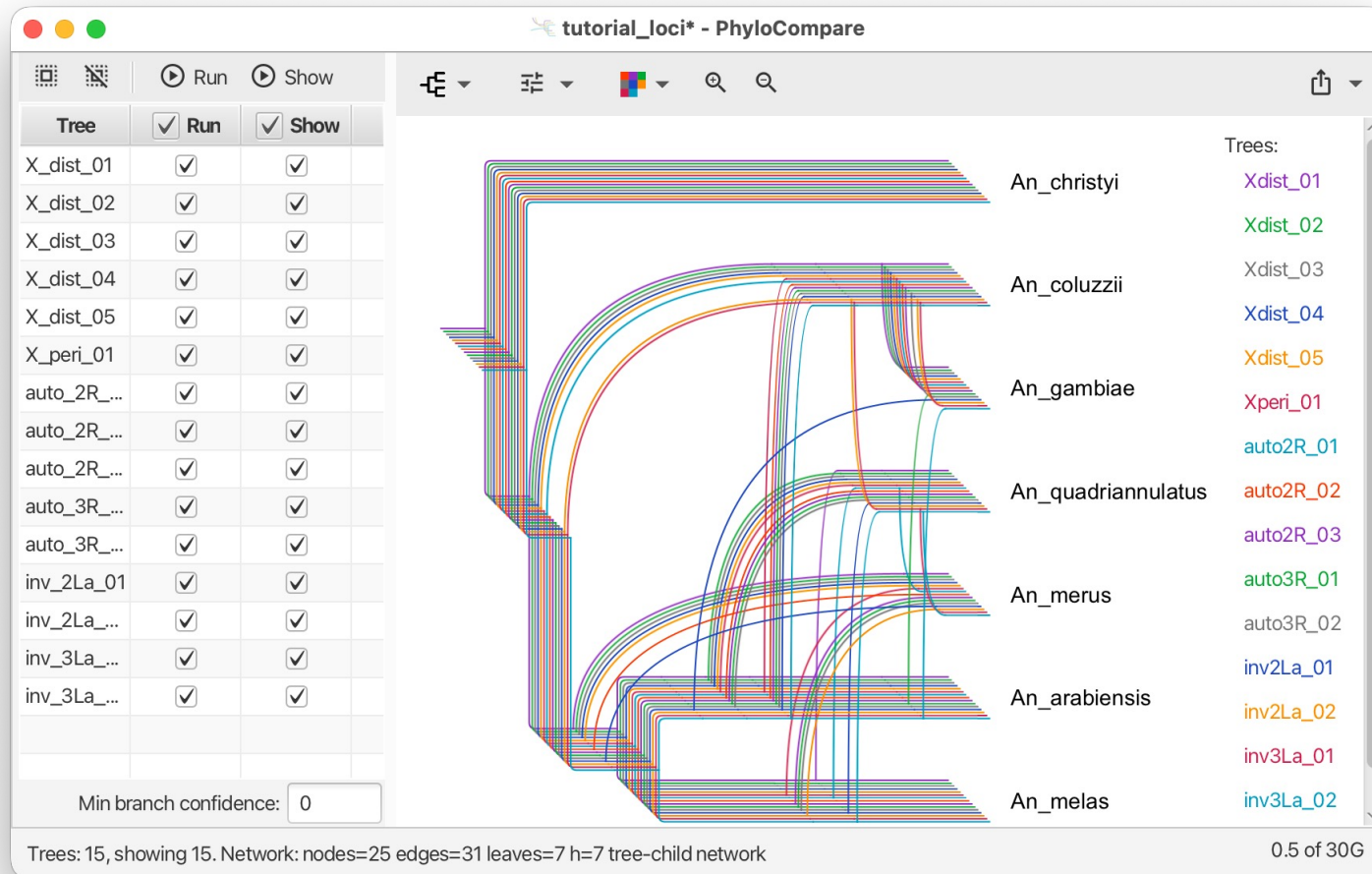




# Hands-on PhyloCompare : from gene trees to a network



# Hands-on PhyloCompare : from gene trees to a network



## How does PhyloFusion's network compare to what you sketched?

- Open your saved sketch from PhyloSketch alongside the PhyloCompare network
- Compare:
  - Same number of reticulations?
  - Same reticulation events (same pairs of lineages involved)?
  - Same direction of gene flow (if your sketch indicated direction)?

## PhyloGuide Q5

**Q:** We have a phylogenetic network for the *Anopheles gambiae* species complex with reticulations between *An. arabiensis* and the (*An. gambiae*, *An. coluzzii*) clade, and between *An. merus* and *An. quadriannulatus*. What does this tell us biologically?

**A:** The reticulations suggest non-tree-like evolution, possibly due to historical gene flow/hybridization or ILS. Different loci may reflect different evolutionary histories across the genome.





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## Back to the biology: what did this tell us about malaria

- **The species branching order:** X chromosome topology is correct; *An. arabiensis* and *An. quadriannulatus* are sister taxa, NOT the three vector species clustering
- **Pervasive autosomal introgression:** especially between *An. arabiensis* and the (*An. gambiae*, *An. coluzzii*) ancestor
- **An unexpected finding** (which we saw too): introgression between *An. merus* and *An. quadriannulatus* in the 3La inversion region
- **The implication for malaria control:** vectorial capacity may have spread between species through introgression - not just inherited vertically. Traits like host preference, insecticide resistance, and desiccation tolerance can move between species

“If introgression moves vector-capability traits, then control strategies that target one species may be partially neutralized by introgression from another.”



# Take-aways from the tutorial

1. **Gene trees disagree systematically across the genome** when introgression is pervasive. This is data, not noise.
2. **Tree-summary methods can be confidently wrong** when the underlying biology is reticulate. ASTRAL got the “wrong” species tree from our data - not because it’s broken, but because its model does not represent introgression.
3. **Networks make the conflict visible**, in three complementary ways: alignment-level (Neighbor-Net), within-locus uncertainty (consensus from posterior), and across-locus reticulation (PhyloCompare).
4. **Hypothesis sketching before automated inference** clarifies what you’re looking for and makes the algorithm’s output interpretable.
5. **AI assistance** (PhyloGuide) is useful for orientation and recommendations but should not replace expert judgment.

# Continuing your phylogenomics journey

- **This tutorial's repository** (everything we did today):  
<https://github.com/husonlab/trees-to-networks-tutorial>
  - Re-runnable on your own data
  - All scripts, alignments, and precomputed outputs available
- **Tools we used today** - links and documentation:
  - SplitsTree: <https://github.com/husonlab/splitstree6>
  - PhyloSketch <https://github.com/husonlab/phylosketch2>
  - PhyloCompare <https://github.com/husonlab/phylocompare>
  - IQ-TREE: <http://www.iqtree.org>
  - BEAST X: <https://beast.community>
  - ASTRAL: <https://github.com/smirarab/ASTRAL>
- **Further reading:** see `References.md` in the repo





# Thank you - and please give us feedback

- Thanks to:
  - All of you for spending the morning with us
  - The Anopheles genomics consortium and Fontaine et al. for the dataset that made this possible
  - The tool developers whose work underlies the tutorial (Minh lab, Suchard lab, Drummond lab, Rambaut lab, Mirarab lab, and our own group)
  - ISCB for the tutorial program

# Share Your Feedback

Scan the QR code to let us know your thoughts on this tutorial.  
Thank you!

